



UNIVERSITÀ DEGLI STUDI
DI GENOVA



Amsterdam UMC

Neuroimaging in Alzheimer's Disease and Related Disorders

What Have We Learned In The Past Year?

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Disclosures



Luigi Lorenzini has Nothing to disclose



Photography is welcomed

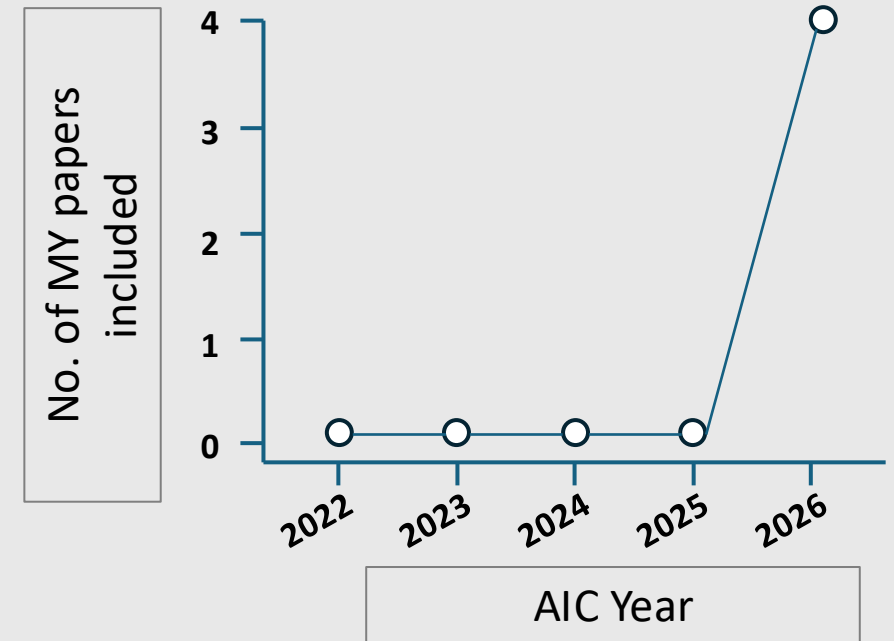
Disclaimers



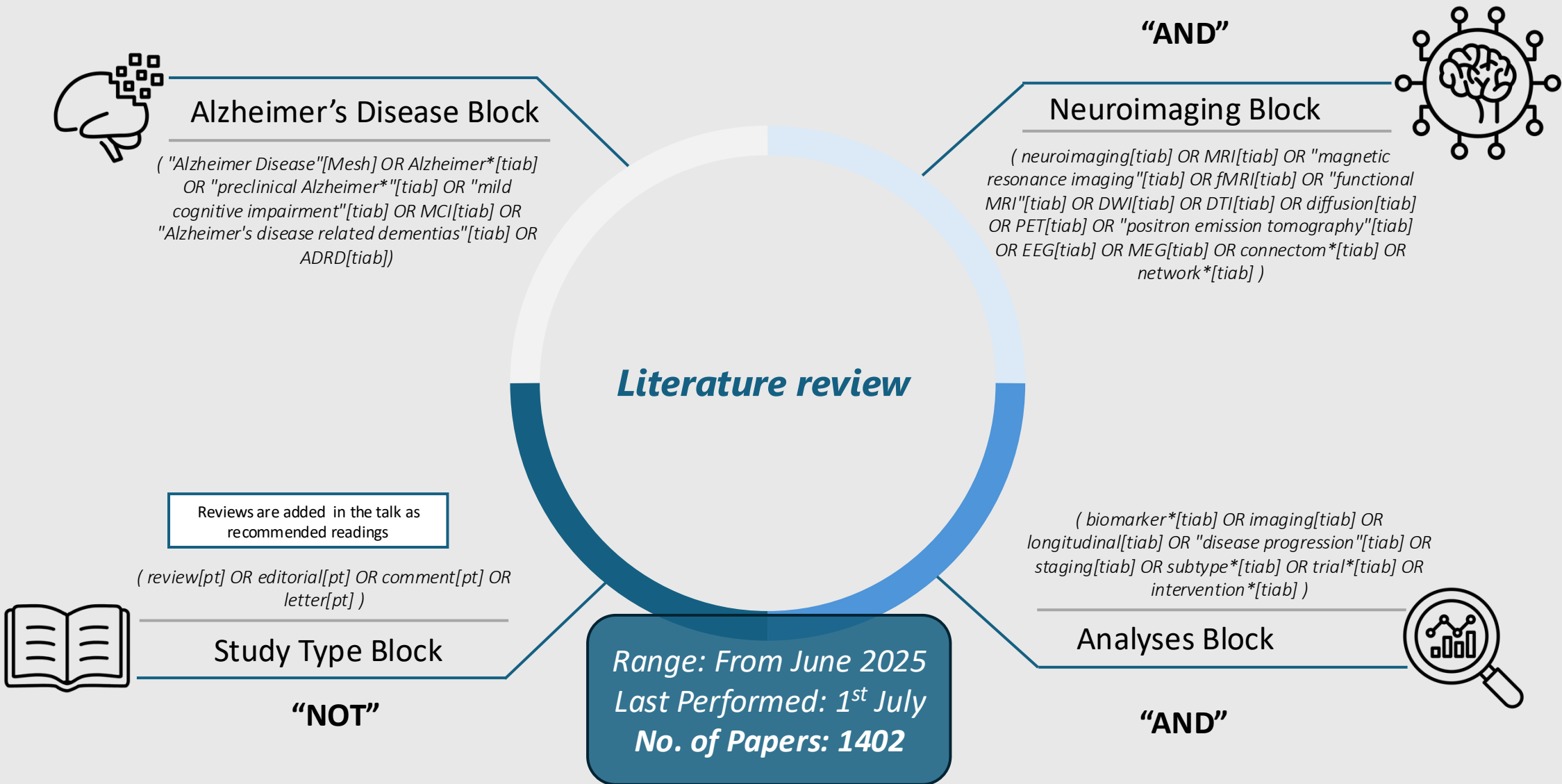
We published a lot!
It was impossible to
cover all themes.



The inclusion of papers
was biased towards my
expertise and
knowledge



Semi-Systematic Papers and Themes Selection



Semi-Systematic Papers and Themes Selection

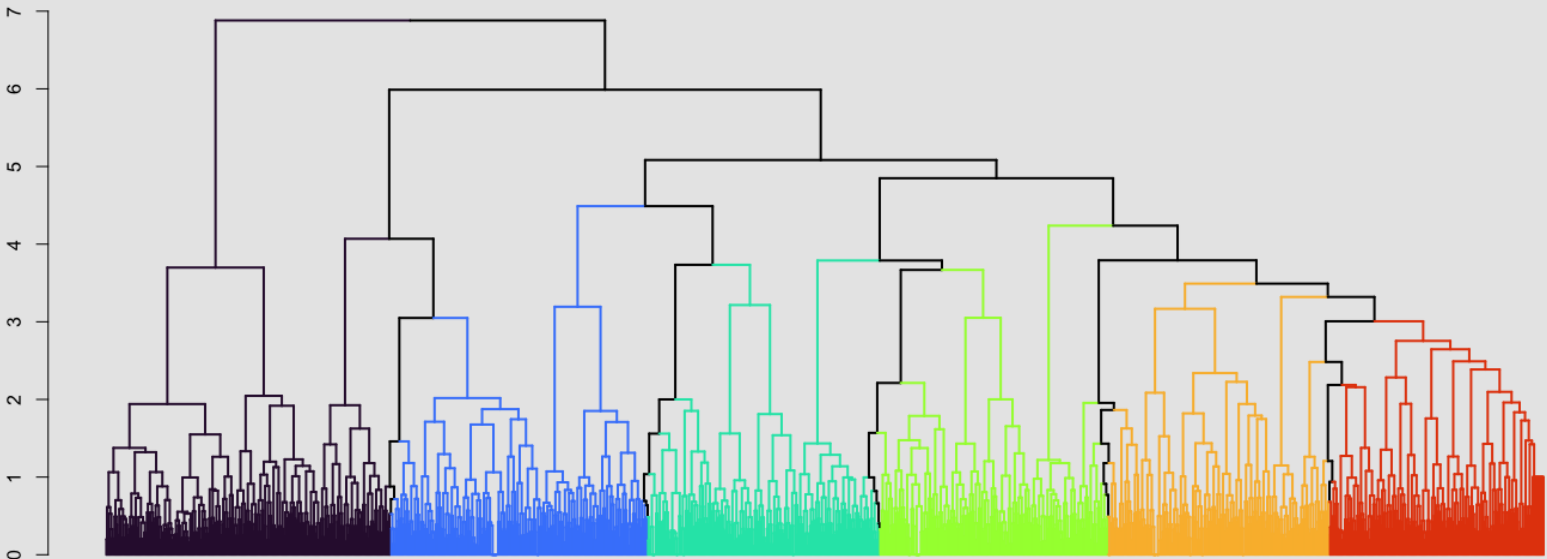
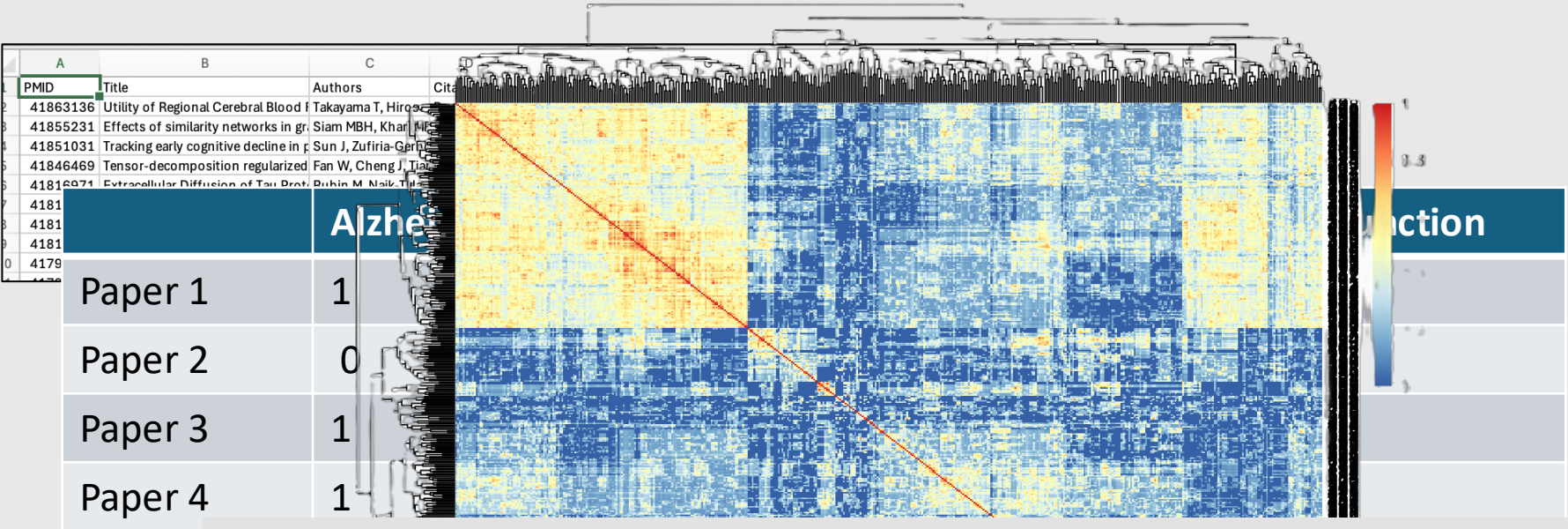
Data-driven Themes Selection

Range: From June 2025
Last Performed: 1st July
No. of Papers: 1402

Document Term Matrix

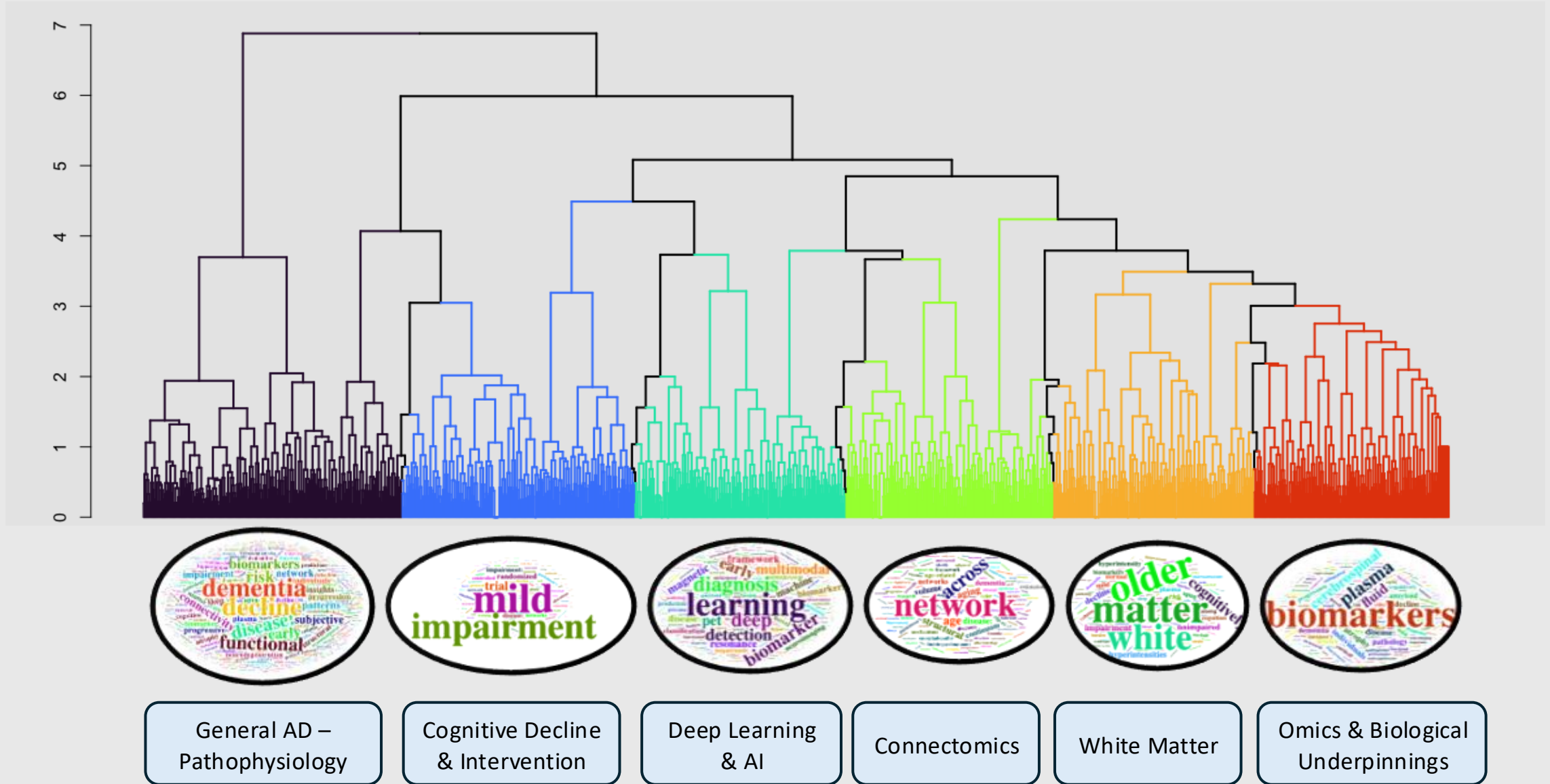
Cosine Similarity Matrix

Hierarchical Clustering

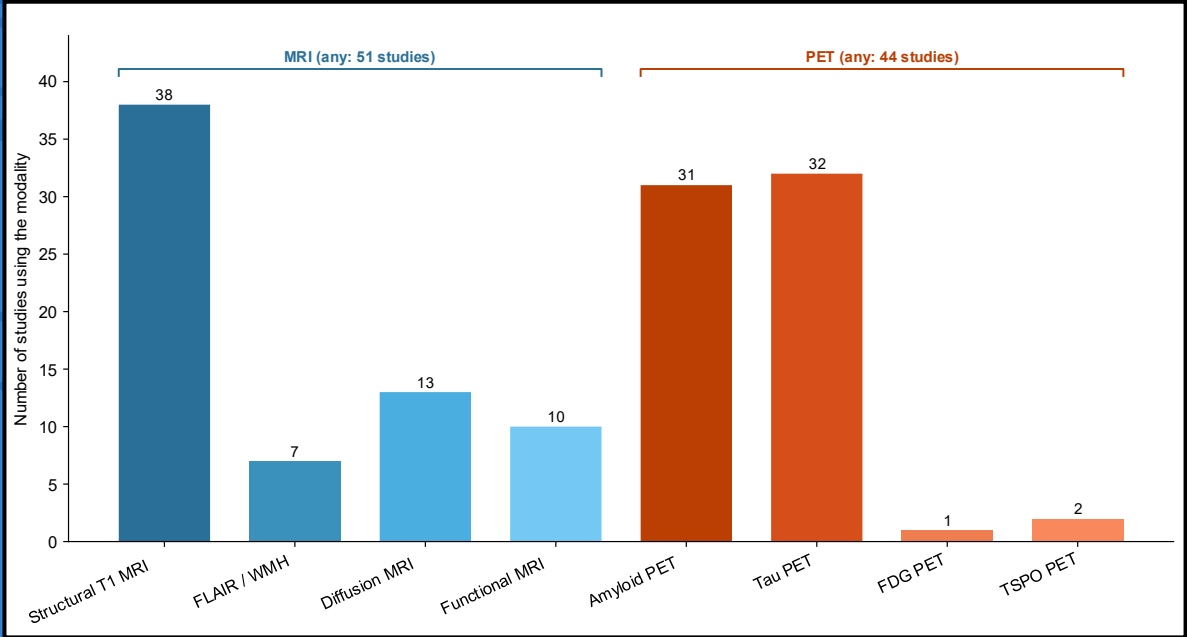
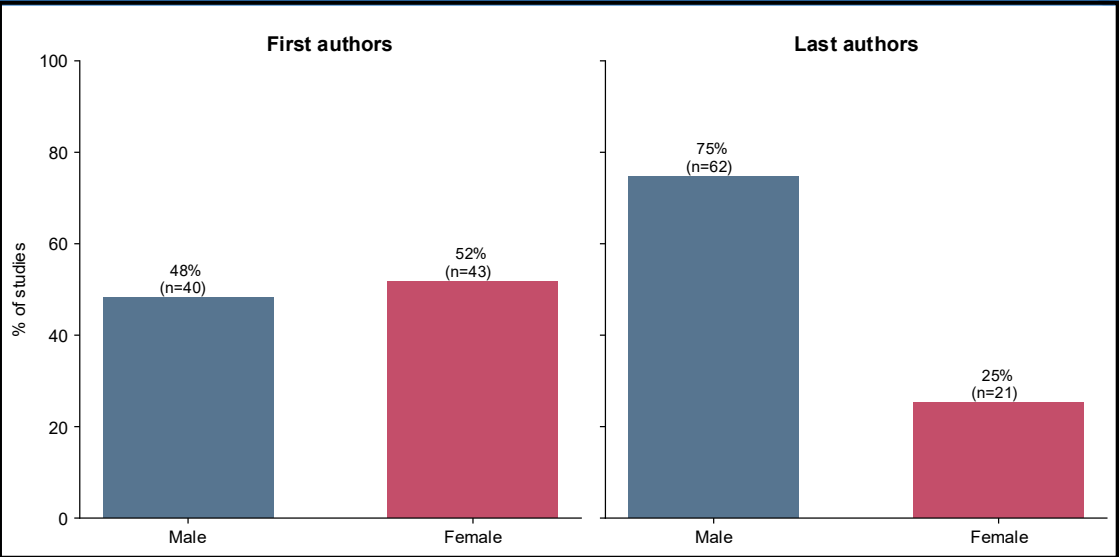
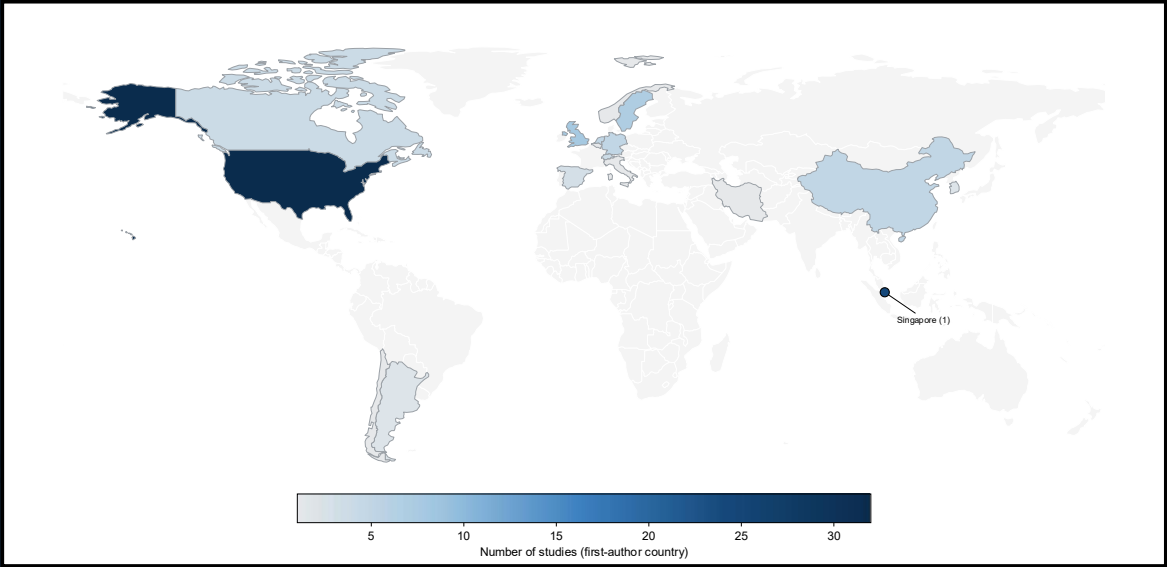
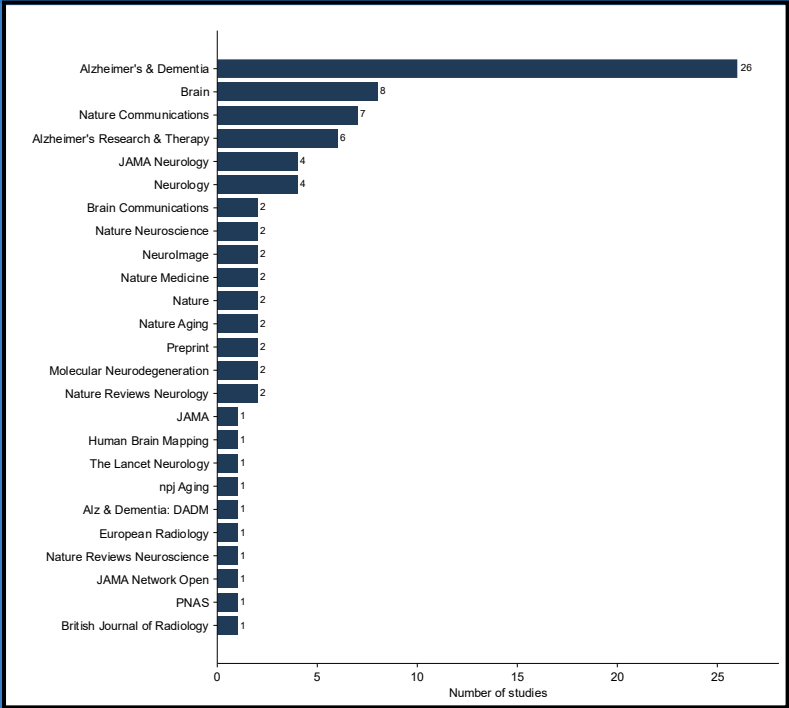


Semi-Systematic Papers and Themes Selection

Data-driven Themes Selection



Semi-Systematic Papers and Themes Selection





General AD –
Pathophysiology



Cognitive Decline
& Intervention



Deep Learning
& AI



Connectomics



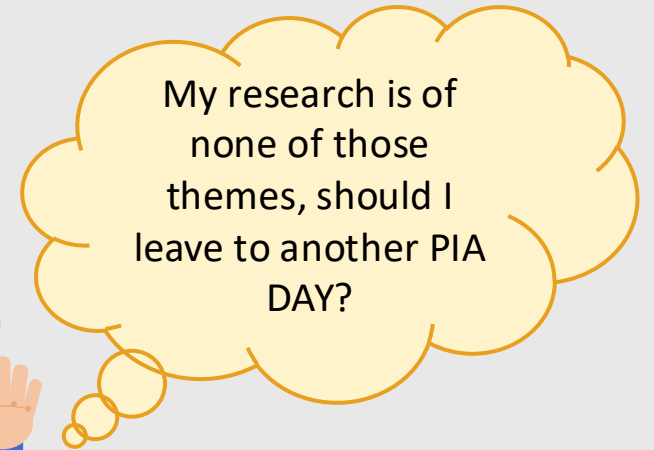
White Matter



Omics & Biological
Underpinnings



+1 Surprise Theme!



Two Transversal Themes

Sex-differences

1

"Female Paradox":
T1 and Tau PET, initial resilience but steeper decline

2

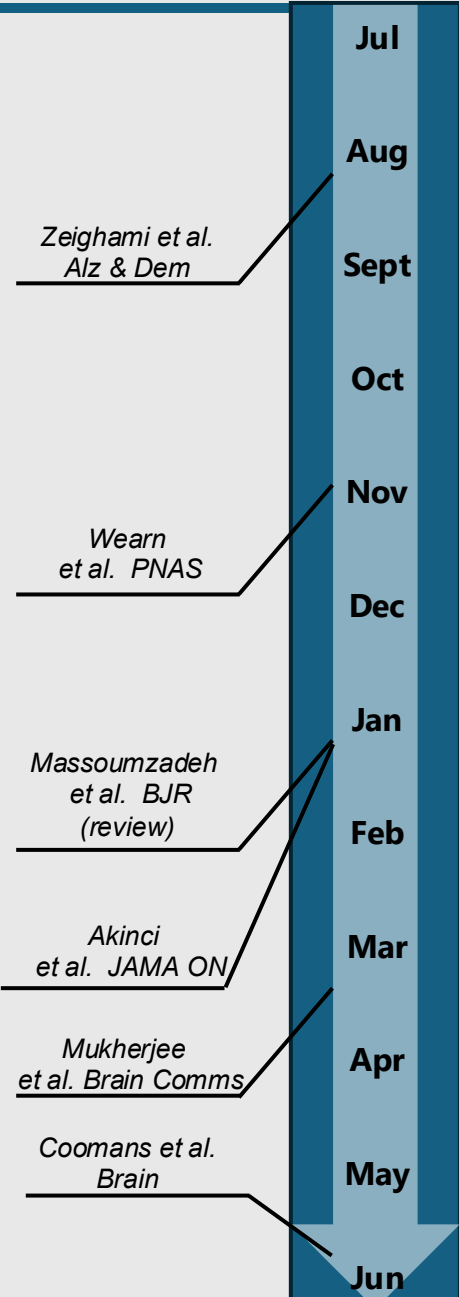
Women show **greater Tau burden** once amyloid is present, and **stronger effect on cognition**

3

Males show **earlier but slower structural grey matter loss** + more brain decline in ageing

4

APOE e4 and Estrogen decline at menopause proposed as mediating mechanisms



Diversity and Accessibility

Two Transversal Themes

Sex-differences

1

"Female Paradox":
T1 and Tau PET, initial resilience but steeper decline

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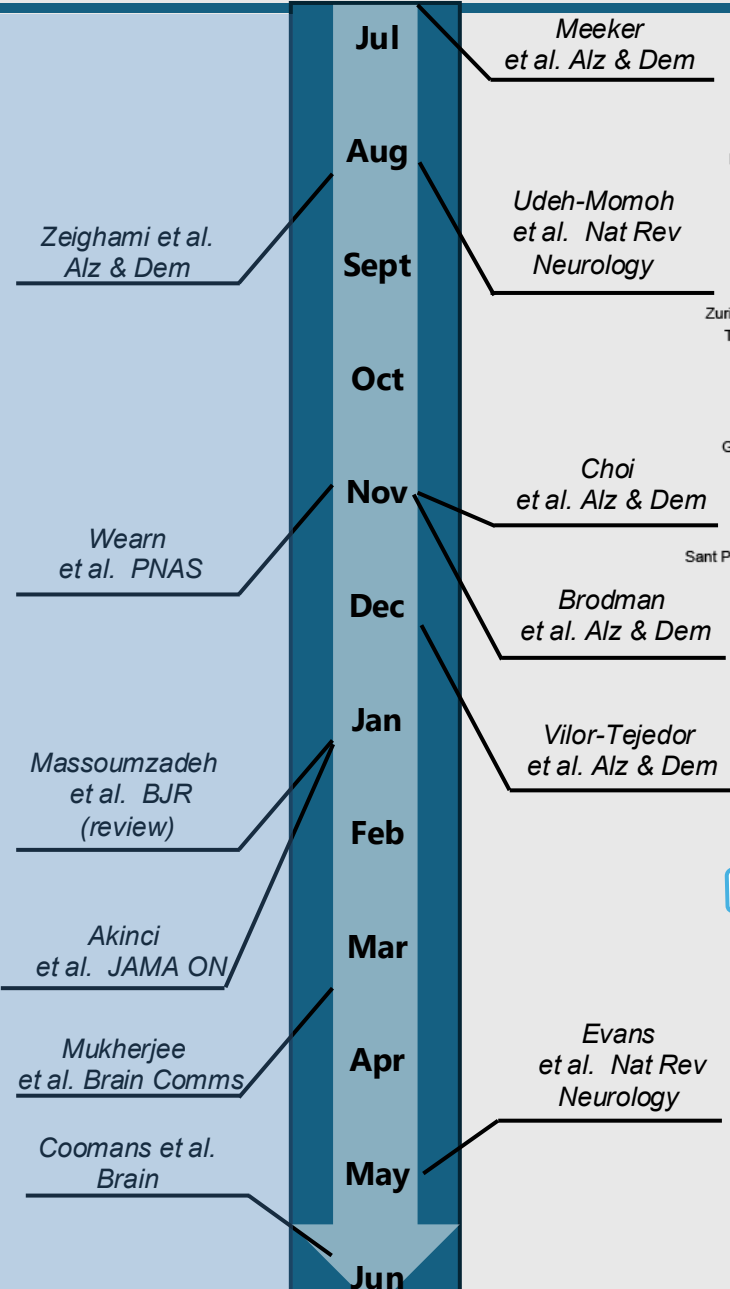
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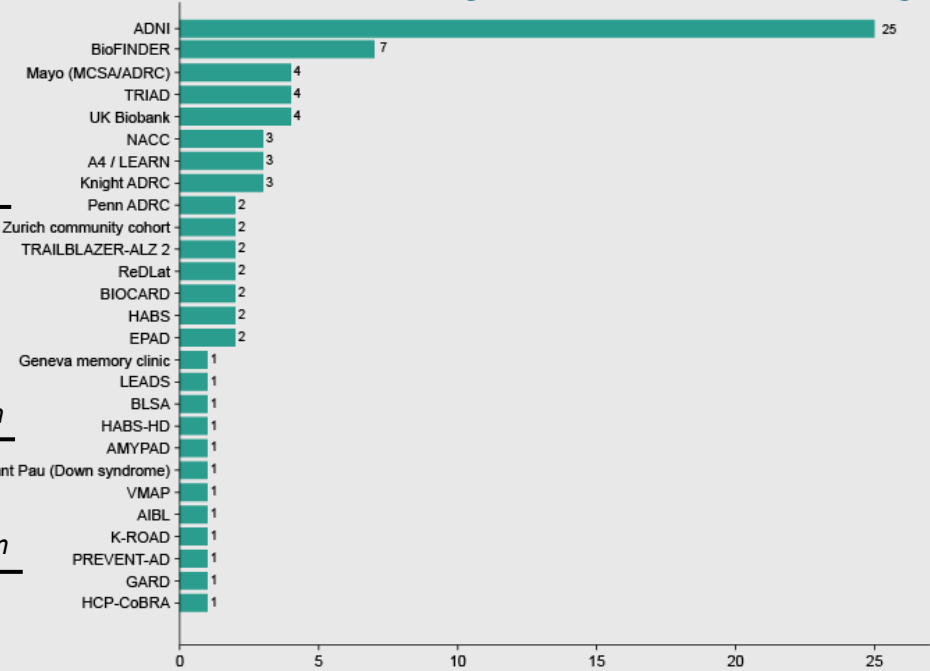
Males show **earlier but slower structural grey matter loss** + more brain decline in ageing

4

APOE e4 and Estrogen decline at menopause proposed as mediating mechanisms



Diversity and Accessibility



1

Disease Frameworks may not generalize
The sequence ATN in AD may differ by race and ethnicity

2

Overcome diversity in health care through **low-field MRI** and **plasma biomarkers**

Two Transversal Themes

Sex-differences

1

“Female Paradox”:
T1 and Tau PET, initial resilience but steeper decline

2

Women show **greater Tau burden** once amyloid is present, and **stronger effect on cognition**

3

Males show **earlier but slower structural grey matter loss** + more brain decline in ageing

4



propagating mediating mechanisms

Meeker et al. Alz & Dem

Udeh-Momoh et al. Nat Rev Neurology

Zeighami et al. Alz & Dem

Choi et al. Alz & Dem

Wearn et al. PNAS

Brodman et al. Alz & Dem

Vilor-Tejedor et al. Alz & Dem

Massoumzadeh et al. BJR (review)

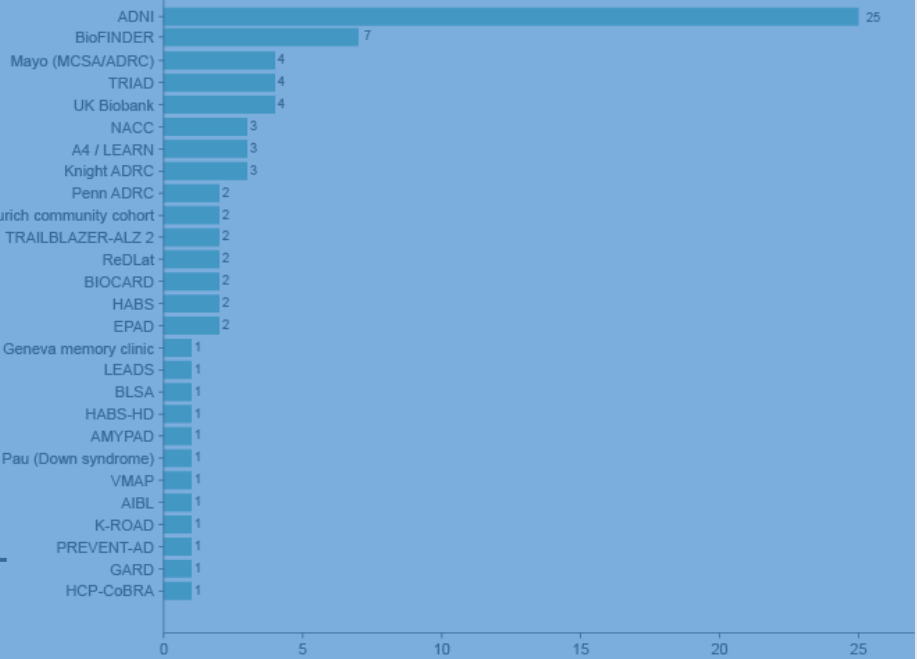
Akinci et al. JAMA ON

Mukherjee et al. Brain Comms

Coomans et al. Brain

Evans et al. Nat Rev Neurology

Diversity and Accessibility



1



2

Overcome diversity in health care through **low-field MRI** and **plasma biomarkers**



General AD – Pathophysiology



Cognitive Decline & Intervention



Deep Learning & AI



Connectomics



White Matter



Omics & Biological Underpinnings



Mismatches

PET Patterns – Neuroinflammation



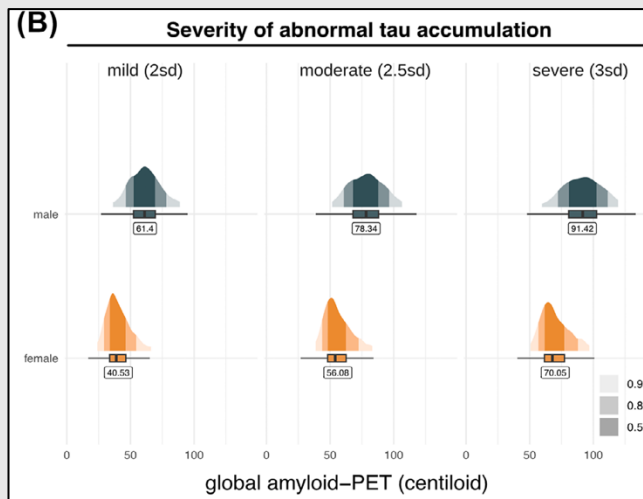
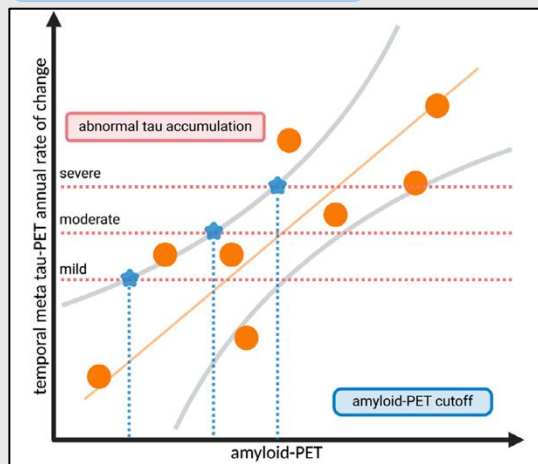
General AD – Pathophysiology

Cutpoints in the AD continuum

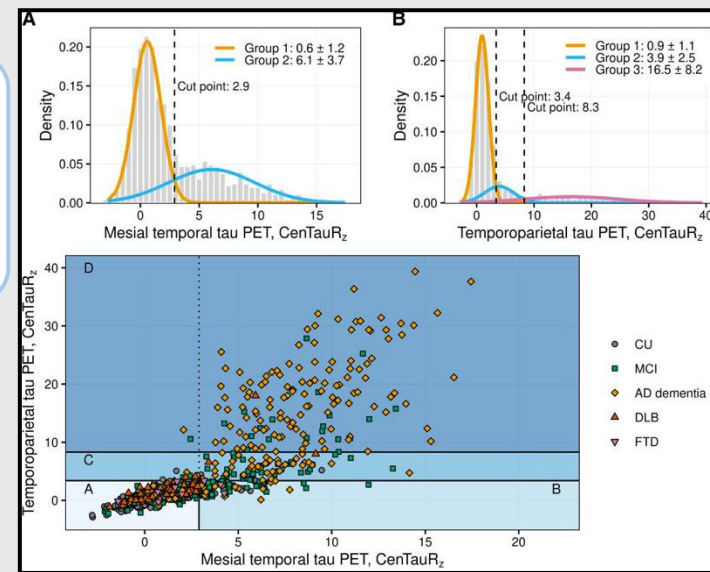
TABLE 3 Biological staging.

	Initial-stage biomarkers (A)	Early-stage biomarkers (B)	Intermediate-stage biomarkers (C)	Advanced-stage biomarkers (D)
PET	Amyloid PET A+T ₂ -	Tau PET medial temporal region A+T ₂ MTL+	Tau PET moderate neocortical uptake A+T ₂ MOD+	Tau PET high neocortical uptake A+T ₂ HIGH+
Core 1 fluid	CSF A β 42/40, p-tau181/A β 42, t-tau/A β 42, and accurate® Core 1 plasma assays can establish that an individual is in biological stage A or higher, but cannot discriminate between PET stages A–D at present.			

Finding CL levels at which amyloid triggers tau

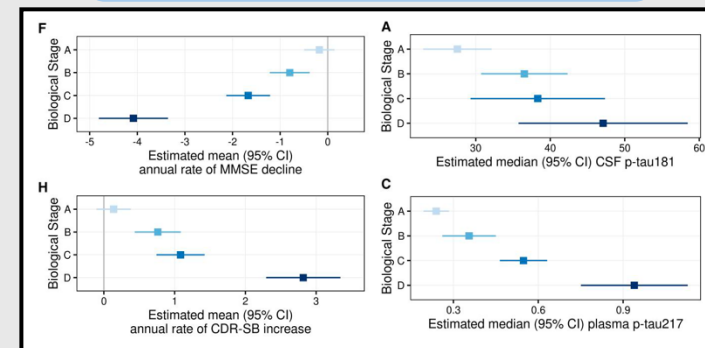


Regional CentTauR cut-offs for biological staging definition



These biological stages capture the increase in clinical severity across the AD spectrum

Women transition to tauopathy at lower amyloid PET levels than men



Jack et al.
Alz & Dem

Johnson et al.
Brain

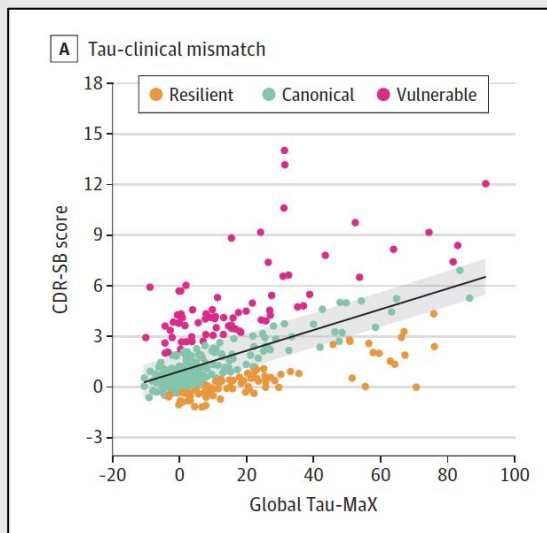
Zhu et al.
Alz & Dem



General AD – Pathophysiology

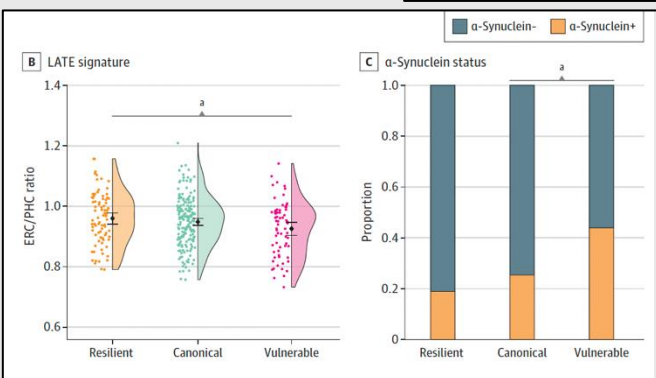
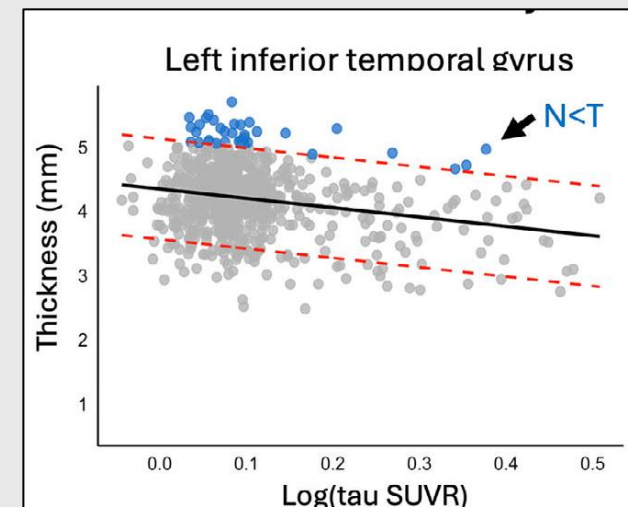
Mismatches

T-C mismatch

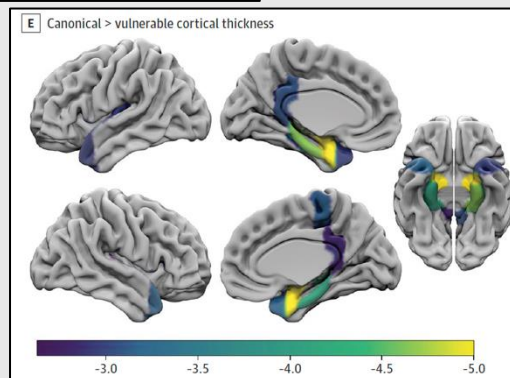


Using the **mismatch** (residuals) between tau either neurodegeneration or clinical to define : Canonical, Vulnerable and Resilient individuals

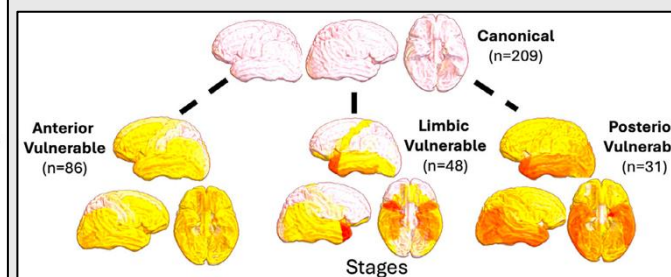
T-N mismatch



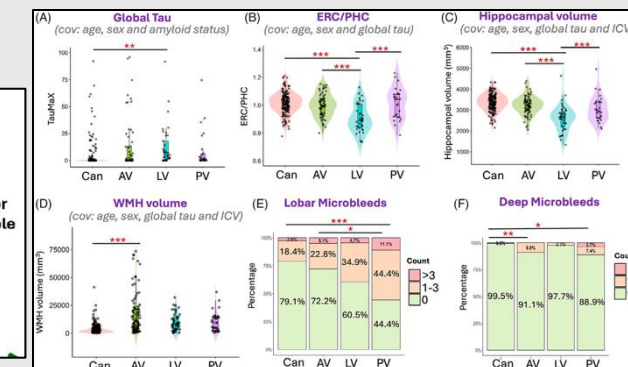
T-C Vulnerable show higher level of co-pathologies



And lower thickness in anterior MTL, temporopolar and fronto-opercular regions.



Three different trajectories (subtypes) of regional vulnerability



Limbic → low ERC/PHC → LATE and TDP-43 co-path
Anterior → high global WMH → Vascular co-path
Posterior → high occipital WMH & MB → CAA co-path

Jack et al.
Alz & Dem

Johnson et al.
Brain



Zhu et al.
Alz & Dem

Brown et al.
Brain

Lyu et al.
Alz & Dem

Lyu et al.
Brain



General AD – Pathophysiology

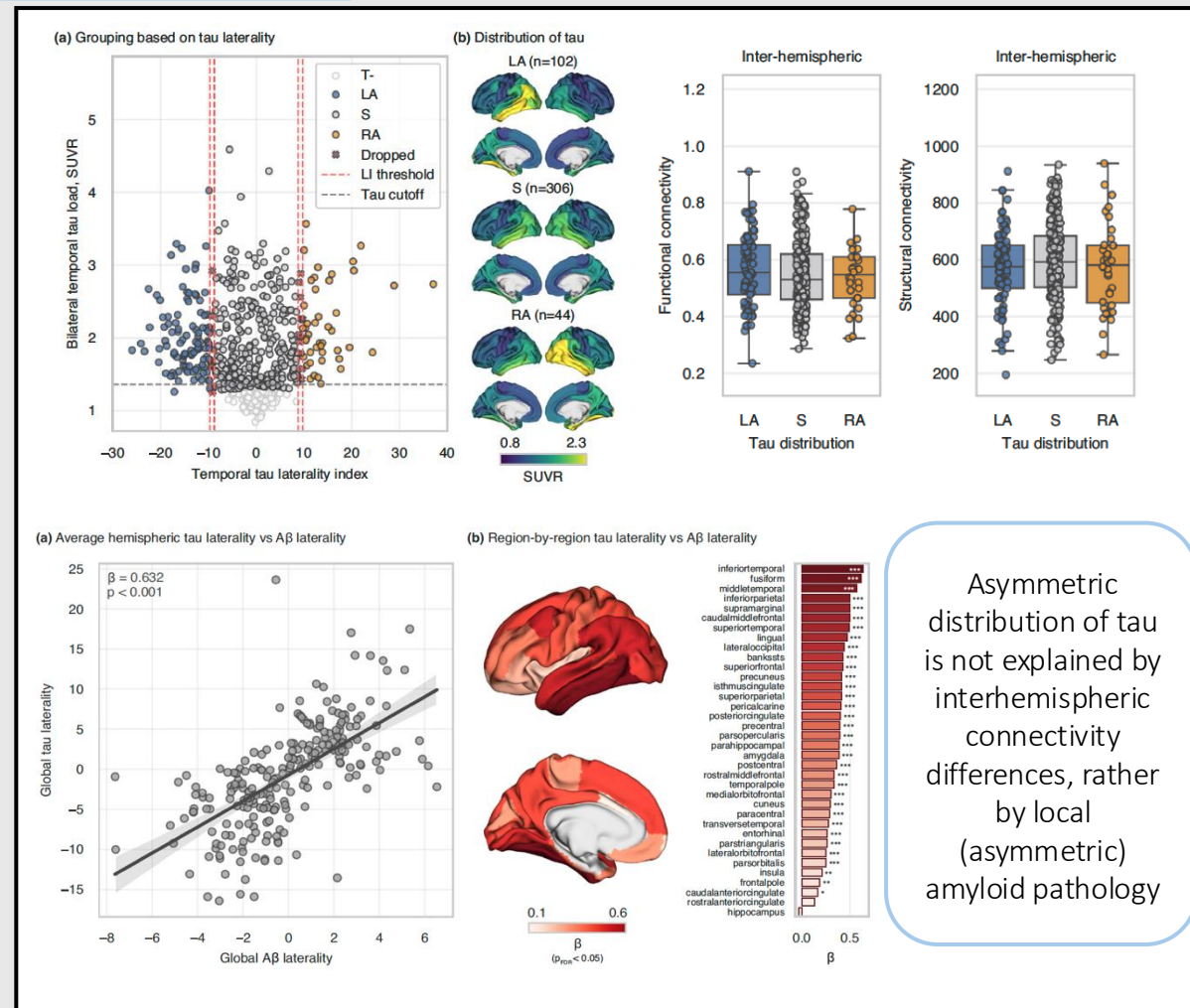
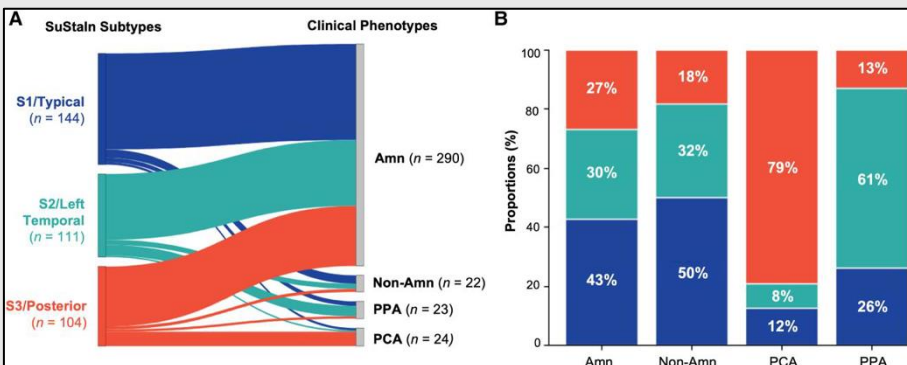
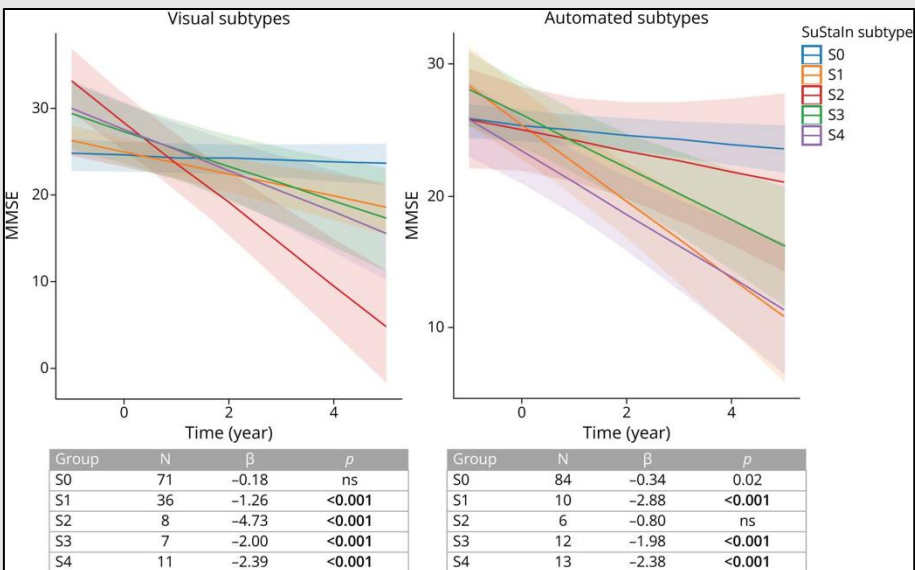
PET Patterns – Amy & Tau

Can tau-PET subtypes be used in clinical practice?

Factoring in asymmetry

Two studies show:

1. Guidelines on visual classification of tau subtypes
2. Good agreement between visual and data-driven subtypes (sustain)
3. Visual subtypes show meaningful difference in longitudinal progression



Anijarv et al.
Nat Comms

Boccalini et al.
Neurology

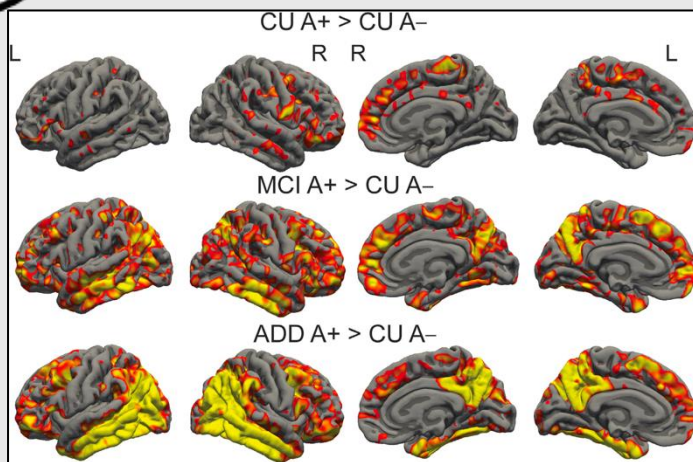
Fonseca et al.
Alz & Dem

Lin et al.
Brain Comms



General AD – Pathophysiology

PET Patterns – Neuroinflammation

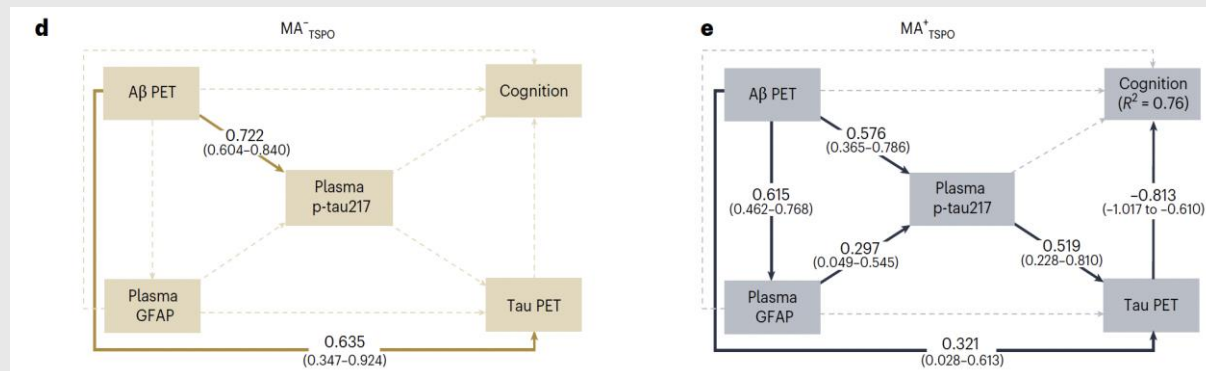
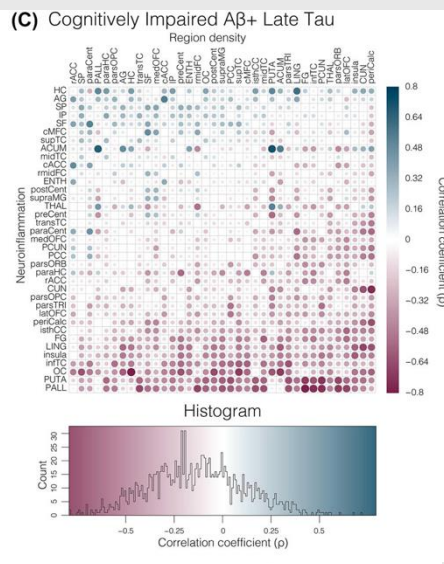
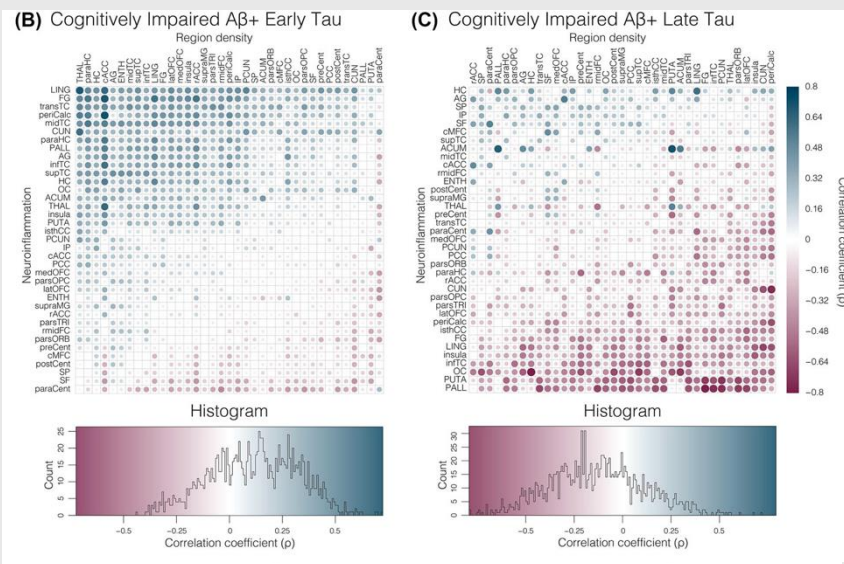
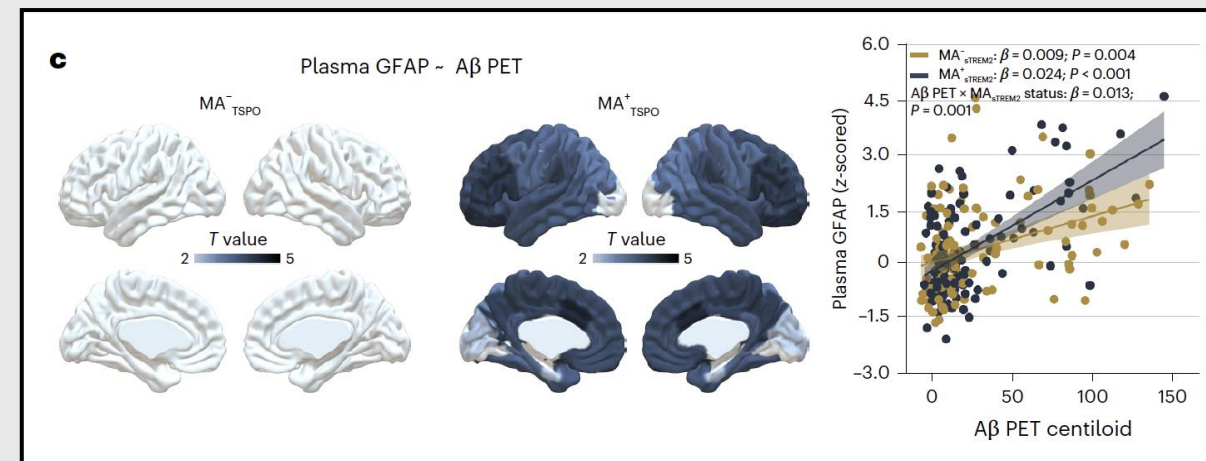


TSPO-PET studies show early inflammation in association with initial amyloid deposition

... this inflammation contributes to early astroglial reactivity...

... and contributes to neurodegeneration in two A β and Tau- related waves...

Inflammatory process may be a gate for initial amyloid deposition to influence subsequent cascade events



Povala et al.
Alz & Dem

Kandimalla
et al. Alz &
Dem~

Ferrari Souza
et al. Nat
Neuro



General AD –
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White Matter



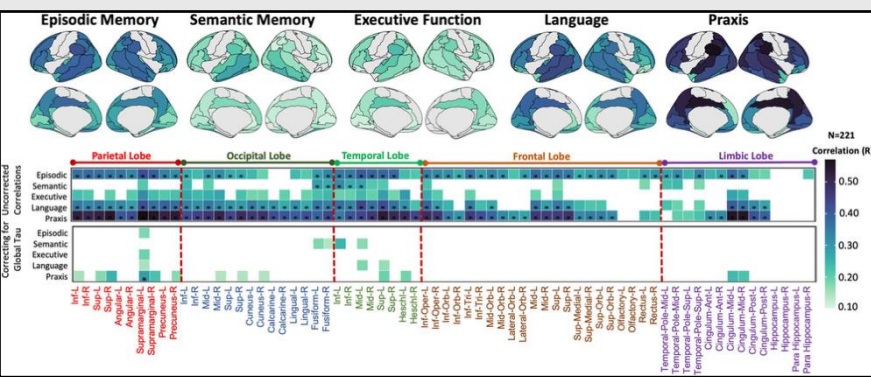
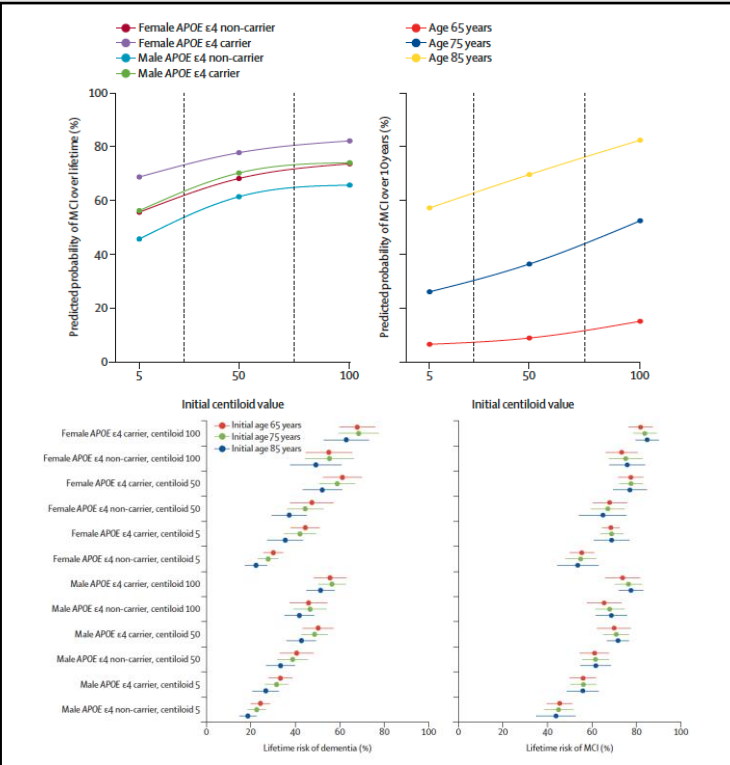
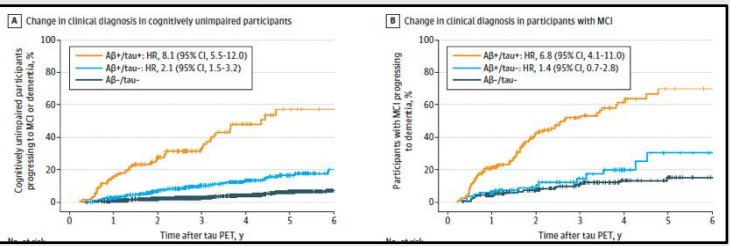
Omics & Biological
Underpinnings



Modeling Cognitive Decline

Clinical Trials – Role of PET and MRI

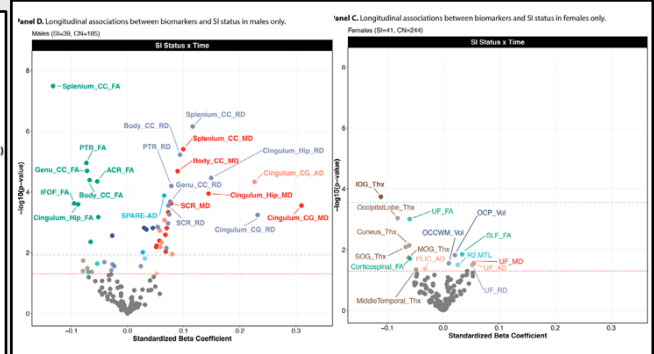
Studies providing estimates of PET positivity associated risk of future clinical outcomes



Spatial patterns of Tau-PET show differential association with domain specific cognitive decline over time

1. Lifetime and 10-year absolute risk of MCI and dementia increase continuously with increasing CL value for all combinations of sex and APOE ε4 status.
2. Amyloid PET CL has the largest effect for lifetime risk of both MCI and dementia.
3. Lifetime and 10-year absolute risk for MCI and dementia is greater for APOE ε4 carriers than non-carriers in all groups.

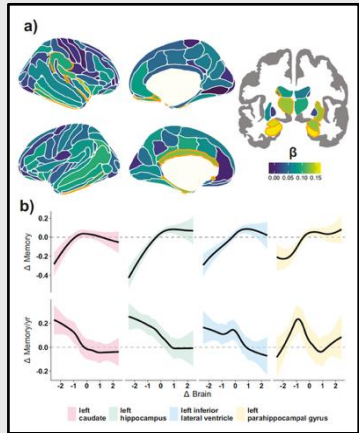
MRI-derived endophenotypes to capture age-related cognitive decline



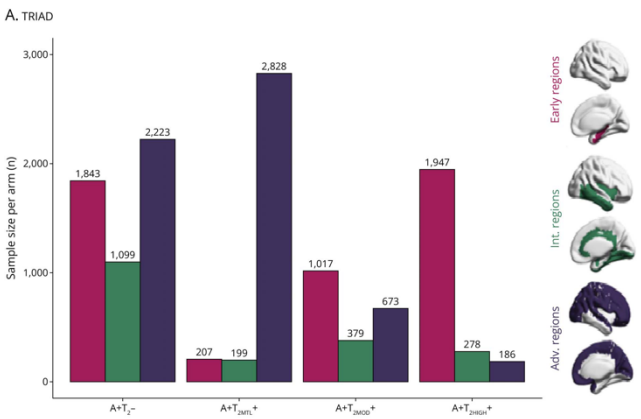
White Matter integrity show the strongest association with subsequent cognitive impairment, especially in males.

- In the **Gray Matter**:
1. Macrostructural (brain volume) properties show concerted brain change memory change longitudinal association in mega-analysis
 2. Microstructural properties (intracellular and dispersed diffusion) show association with long cognition

	R ²		
	0.110	0.670	
	Intracellular Dispersion	Free	
Dorsal attention	0.225	0.121	0.656
Default mode	0.323	0.001	0.665
Frontoparietal	0.266	0.001	0.648
Limbic	0.137	0.110	0.430
Sensorimotor	0.312	0.151	0.670
Subcortical	0.303	0.020	0.496
Ventral attention	0.364	0.011	0.665
Visual	0.134	0.128	0.597

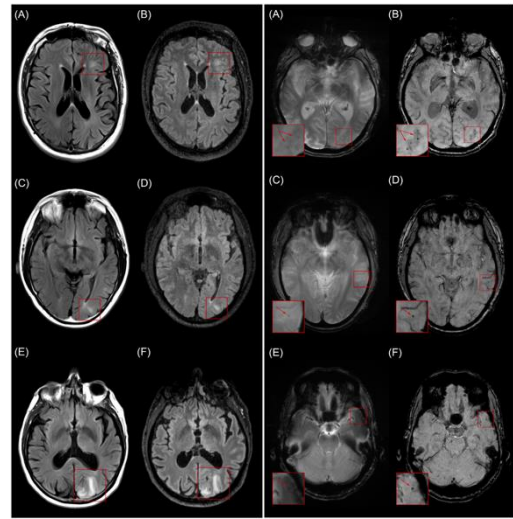
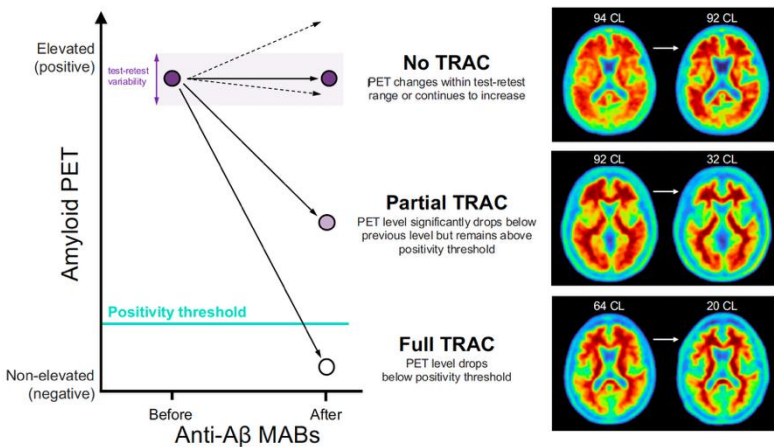


STRATIFY & PLAN

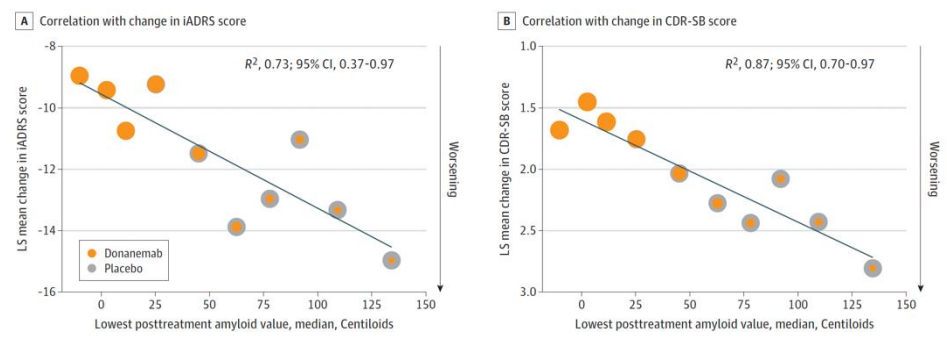


- Combining outcome regions selection and baseline biological staging reduced sample size in clinical trials
- Amyloid-PET Treatment Related Amyloid Clearance (TRAC) provides a dynamic biomarker framework to classify patients as they progress in clinical trials and their BM status changes
- Accelerated 3D MRI achieve comparable performance to gold standard in detecting ARIA-E and drastically reduces scan time

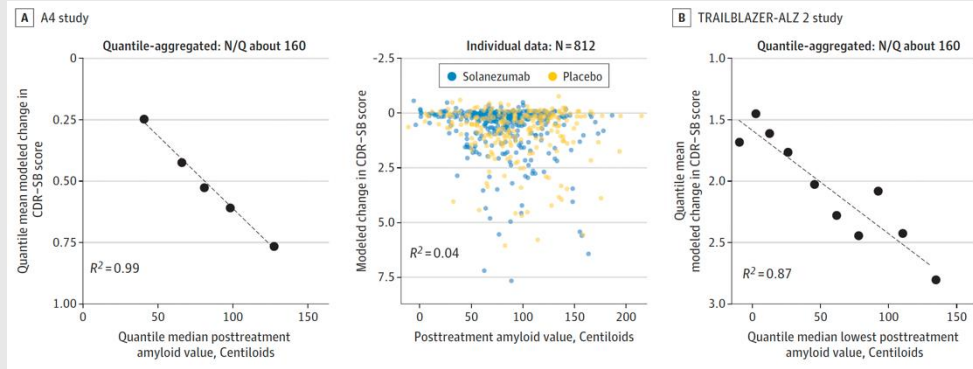
MONITOR : Biomarker Change



RE-ANALYSE



Re-analysis of TRAILBLAZER-AL2 Donanemab data shows association between post-treatment amyloid levels and clinical outcomes





General AD – Pathophysiology



Cognitive Decline & Intervention



Deep Learning & AI



Connectomics



White Matter



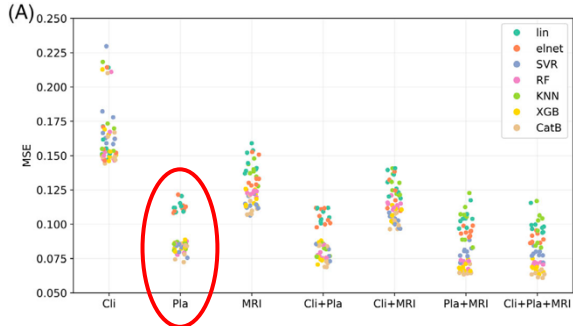
Omics & Biological Underpinnings



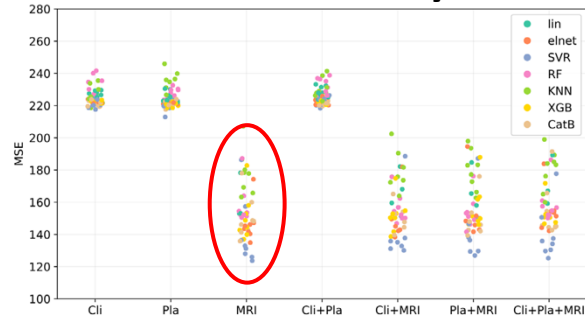
Deep Learning & AI

Biomarkers accessibility: Predicting PET

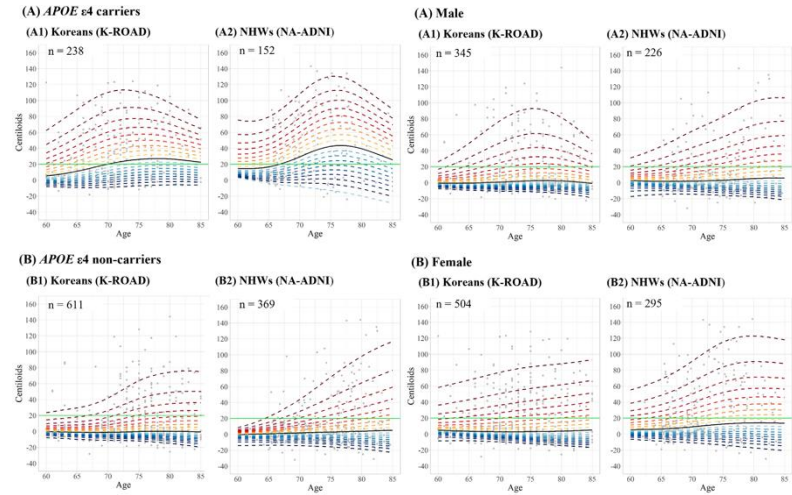
Tau Load



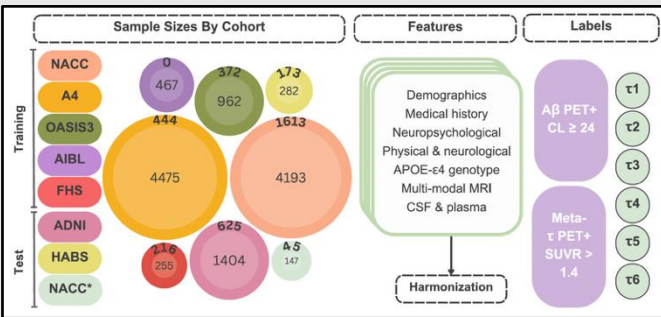
Tau Laterality



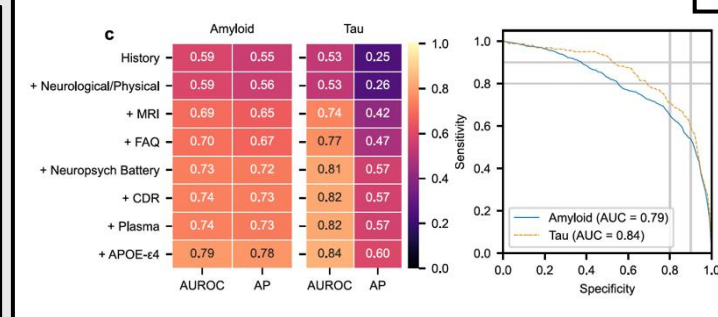
Modelling disease progression and heterogeneity



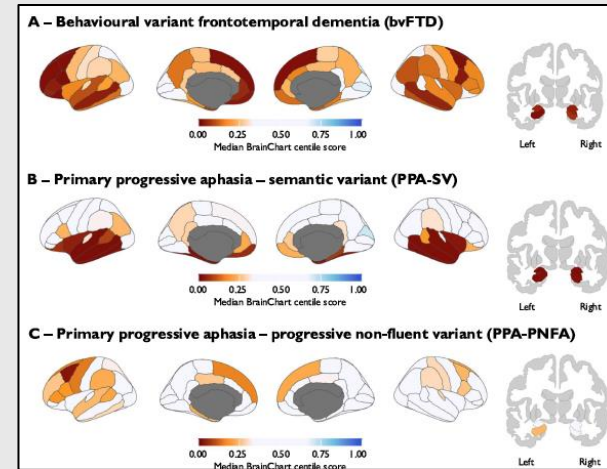
GAMLSS models describe the age-related $A\beta$ trajectories, stratified by sex, APOE and ethnicity



Transformer-based machine learning model on a large diverse cohort with multimodal to predict individual status on amyloid and tau PET



Achieved strong performance on predicting PET status on external data, shows importance of data types in the prediction, and closely match findings from post-mortem data



Normative modelling of brain volumes, or Brain Charts, application in clinical settings help distinguishing between different disease variants.

Bethlehem et al. Nature

Karlsson et al. Alz & Dem

Jasodanand et al. Nat Comms

Lee et al. Alz & Dem

Parker et al. Alz Res & Therapy

Gong et al. Mol Neur.

Chen et al. Eur Rad

Trelle et al. Mol Neur.



General AD – Pathophysiology



Cognitive Decline & Intervention



Deep Learning & AI



Connectomics



White Matter



Omics & Biological Underpinnings



An update on the role of FC

Adding layers

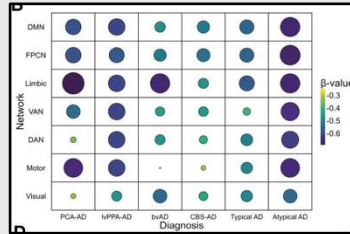


Connectomics

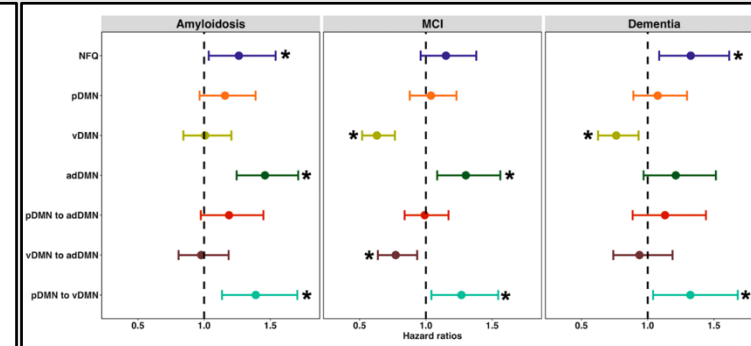
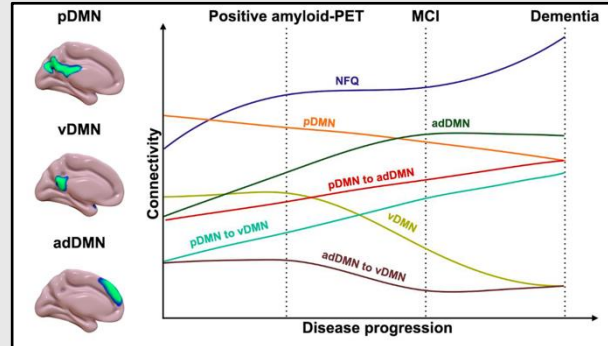
An update on the role of FC

FC mediated Tau progression

The FC mediated hypothesis of tau propagation from local epicenters confirmed for all clinical variants of AD



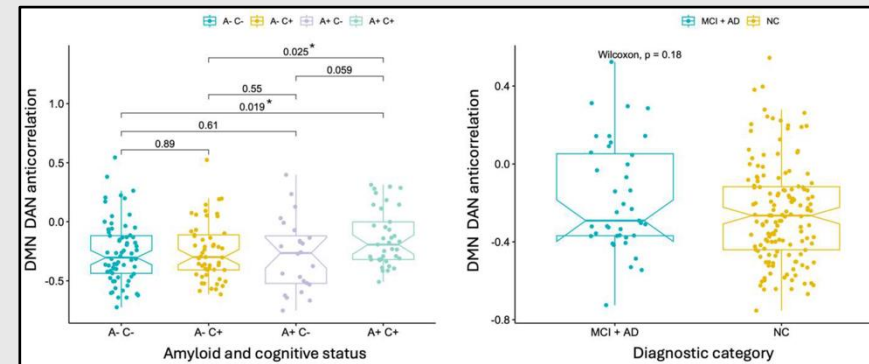
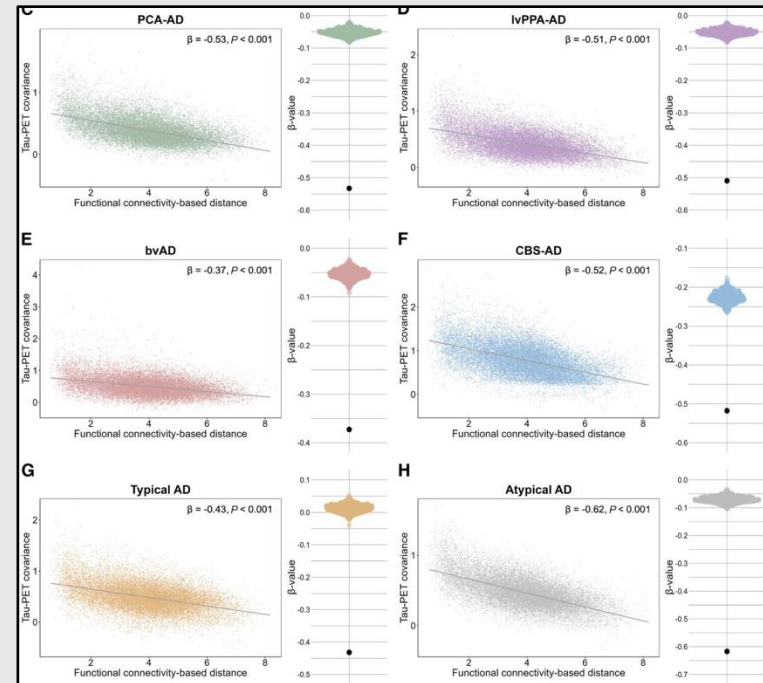
Network Failure Cascade



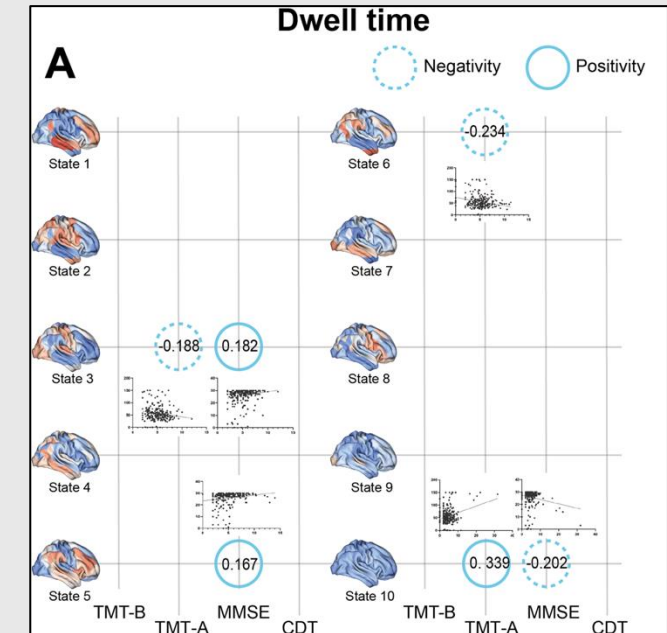
Confirm the network failure hypothesis, with

- decreases in within posterior DMN and
- increases in posterior-to-other DMN subsystem connectivity

Predict future amyloidosis, MCI and dementia



In more advanced stages, reduced interaction of the DMN with other networks (DAN) is particularly related to the onset of cognitive impairment, also confirmed by dynamic connectivity analysis



De Bruin
et al. Brain

Lombardo et al.
Neuroimage

Shi et al. Alz
Res & Ther

Cha et al. Alz &
Dem



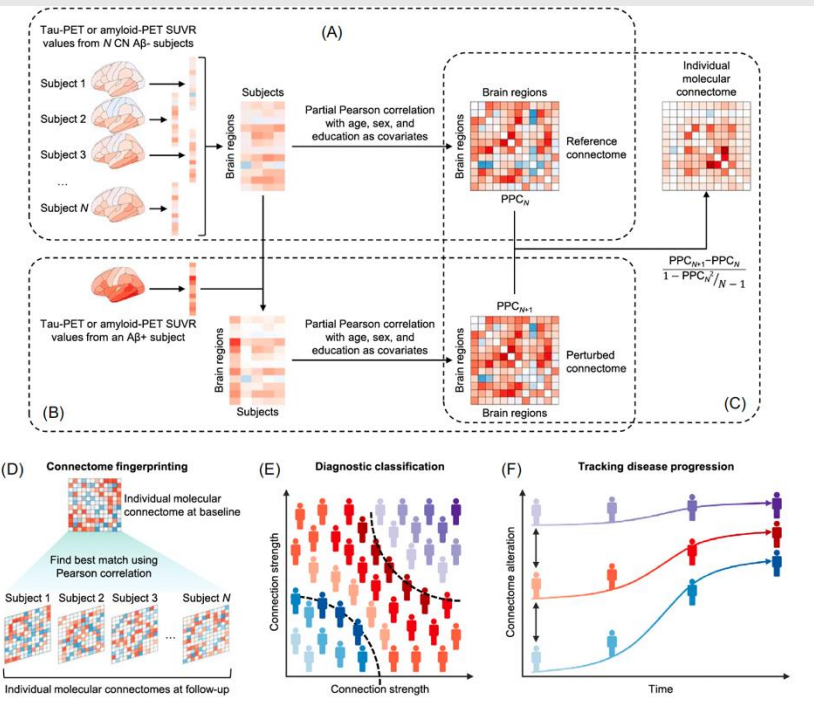
Corriveau-
Lecavalier
et al. Brain



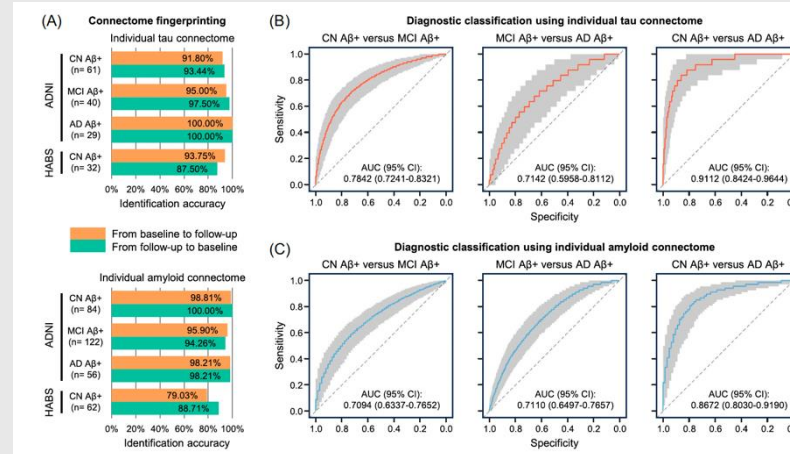
Connectomics

Adding layers

Perturbation approach to compute connectivity from single subject acquisitions of PET (amy or tau) and T1w

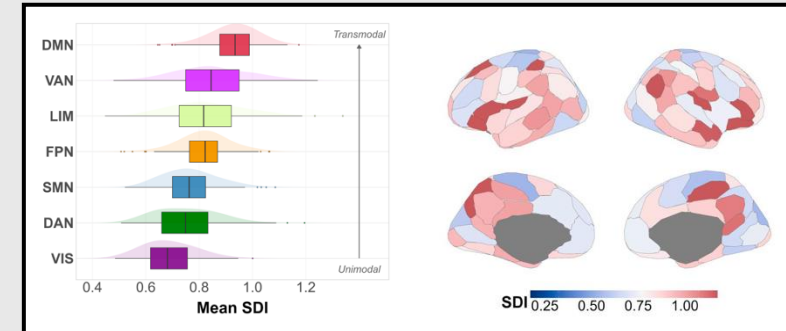


Molecular (PET) connectivity

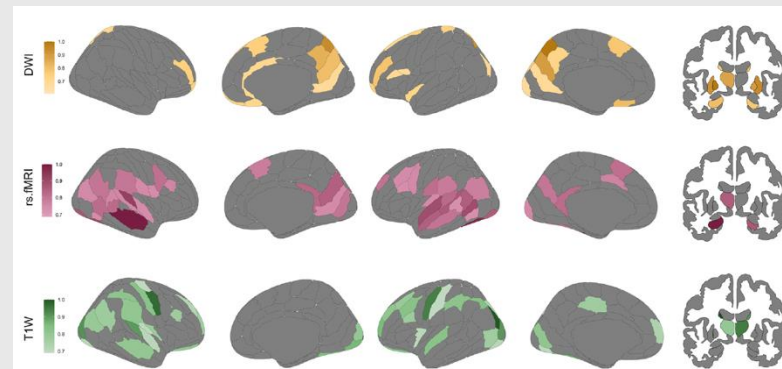


Putting layers together

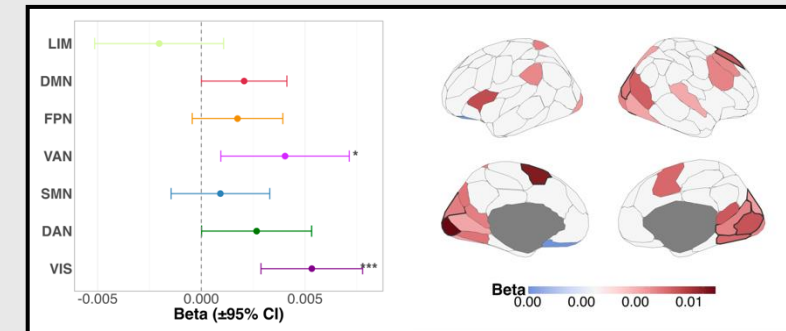
Structural functional (de)coupling in cognitively unimpaired individuals



Gray Matter (T1w) connectivity



Amyloid promotes S-F decoupling in posterior and temporal regions



PET connectivity: Improves PET-only prediction of cognitive decline and diagnosis conversion

T1w connectivity: outperformed anatomical and functional connectivity in predicting age and cognition, showing the earliest age-related changes across the adult lifespan.

Brown et al. Nat Comms

Montagnese et al. Alz & Dem

Xu et al. Alz & Dem

Zufiria-Gerbolés et al. npj Aging

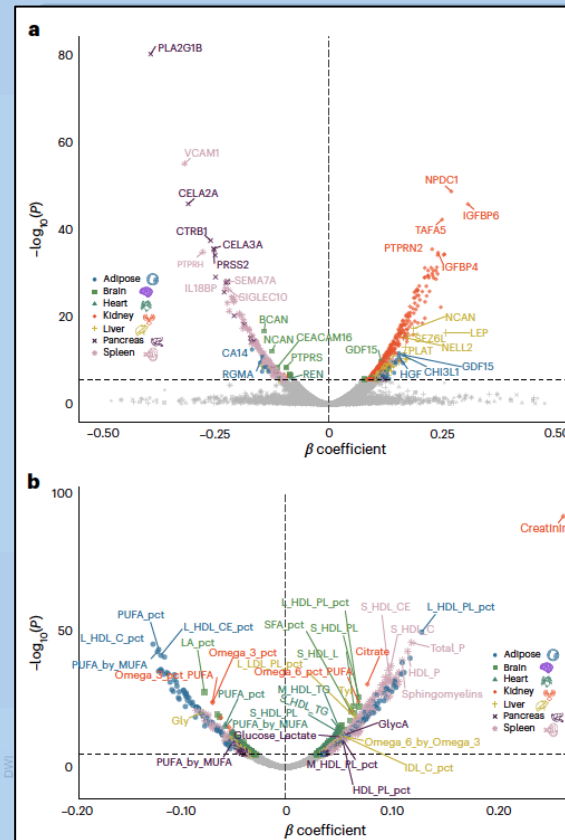
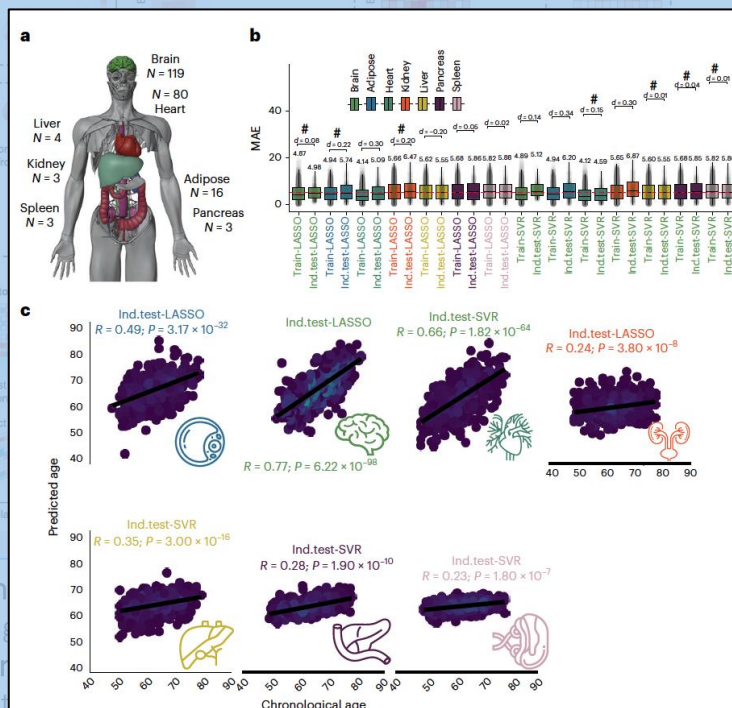
Arunachalam et al. Comms Med



Connectomics

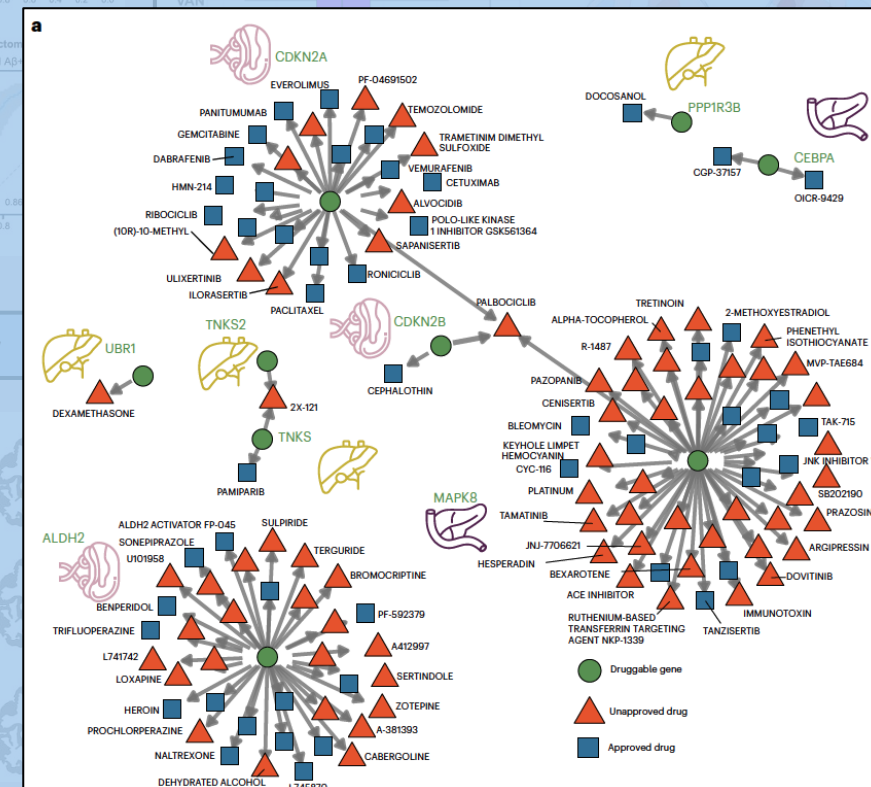
Adding layers - Beyond the Brain

Following the previous work on phenoBAGs, a new model for 7 MRI derived biological age gaps:
the brain, the heart, the liver, adipose tissue, spleen, kidney and pancreas



Identified proteomics, metabolomics, and genetic signature related to MRI BAGs, showing both organ specific and systemic associations

identify 9 druggable genes that are involved in these MRI bags, and show that there are drugs that act on them and could be repurposed to address systemic ageing



Wen et al.
Nat Comms

Cao et al. Nat
Med

Montagnese et
al. Alz & Dem

Xu et al. Alz &
Dem

Zufiria-Gerbolés
et al. npj Aging

Arunachalam et
al. Comms Med



General AD –
Pathophysiology



Cognitive Decline
& Intervention



Deep Learning
& AI



Connectomics



White Matter



Omics & Biological
Underpinnings



The mechanistic role of WM in AD



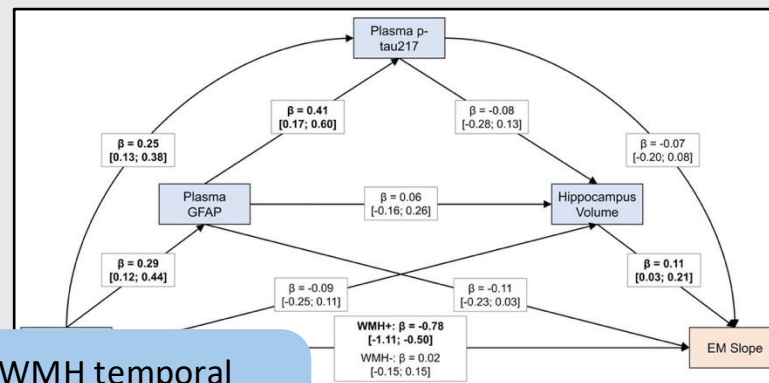
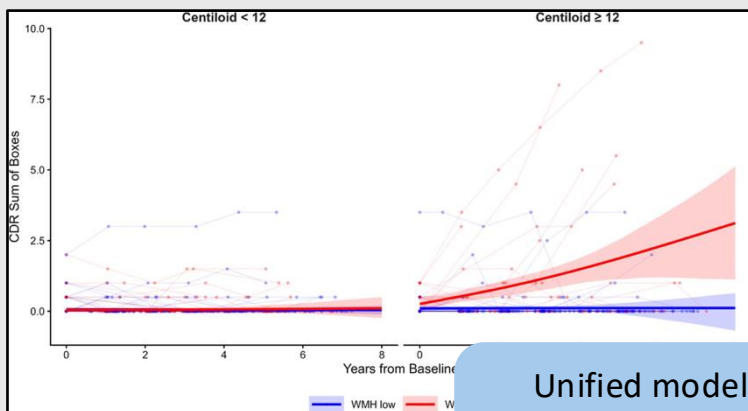
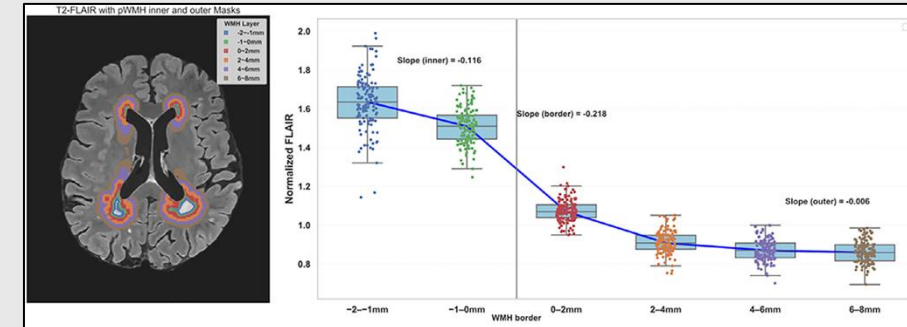
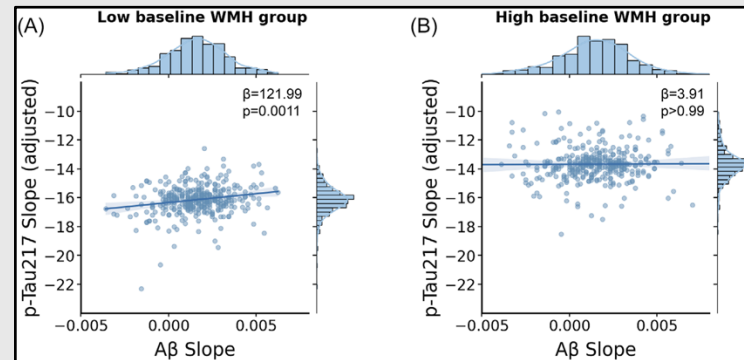
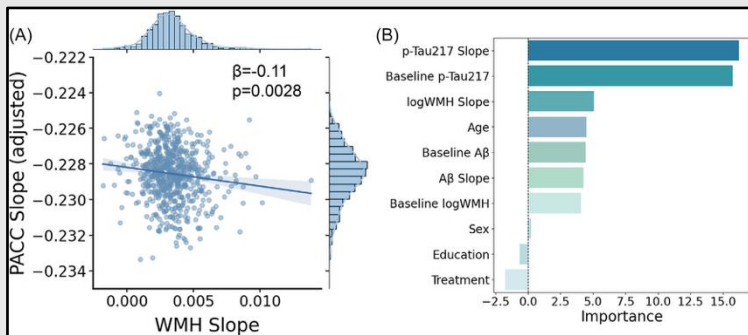
White Matter

White Matter Hyperintensities – An update

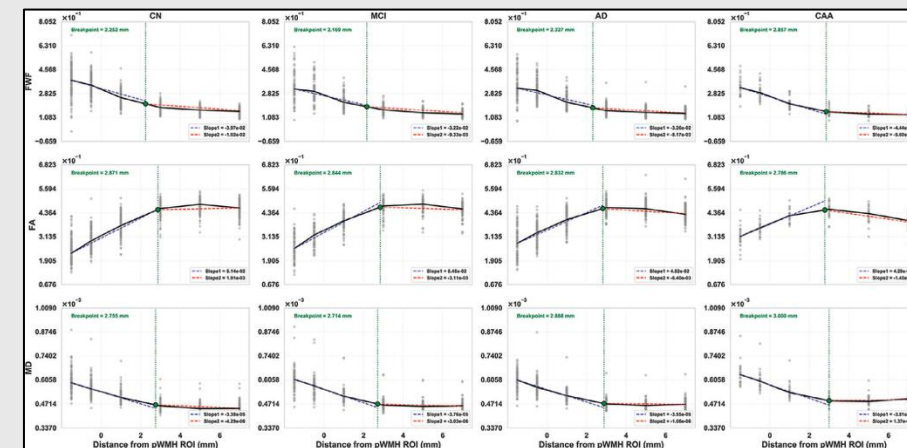
WMH contribute to cognitive decline and interact with other pathological processes in promoting cognitive decline

However, results on whether WMH modulates subsequent pathological events are inconsistent

Perilesional microstructural characterization provide information about underlying pathology:
CAA shows a distinctive pattern of FW changes in perilesional regions with more distal breakpoints and slower plateauing



Unified model of WMH temporal influence and interactions with BM across the AD continuum?



Altschuler et al.
Alz Res &
Therapy

Lin et al. Alz &
Dem

Yang et al. Alz
& Dem

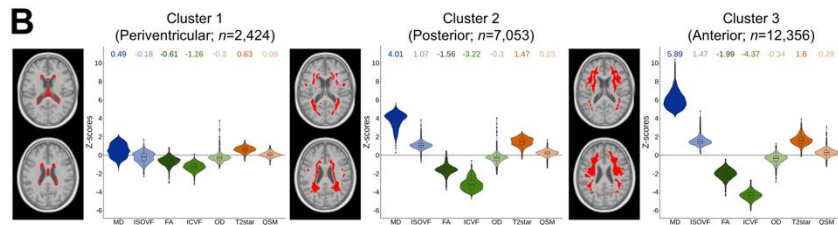
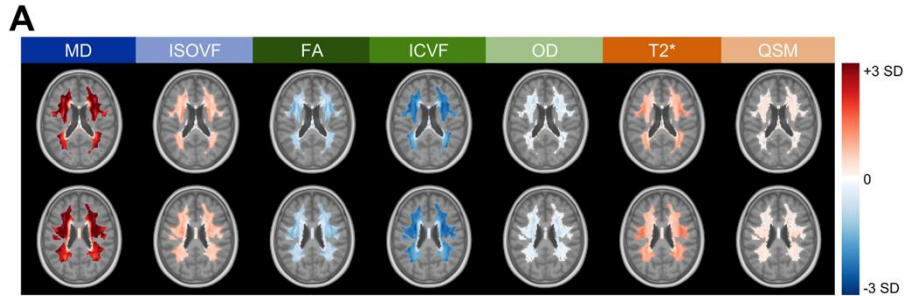
Morillo-Nieto
et al. Alz & Dem

Bachmann et
al. Alz Res &
Therapy



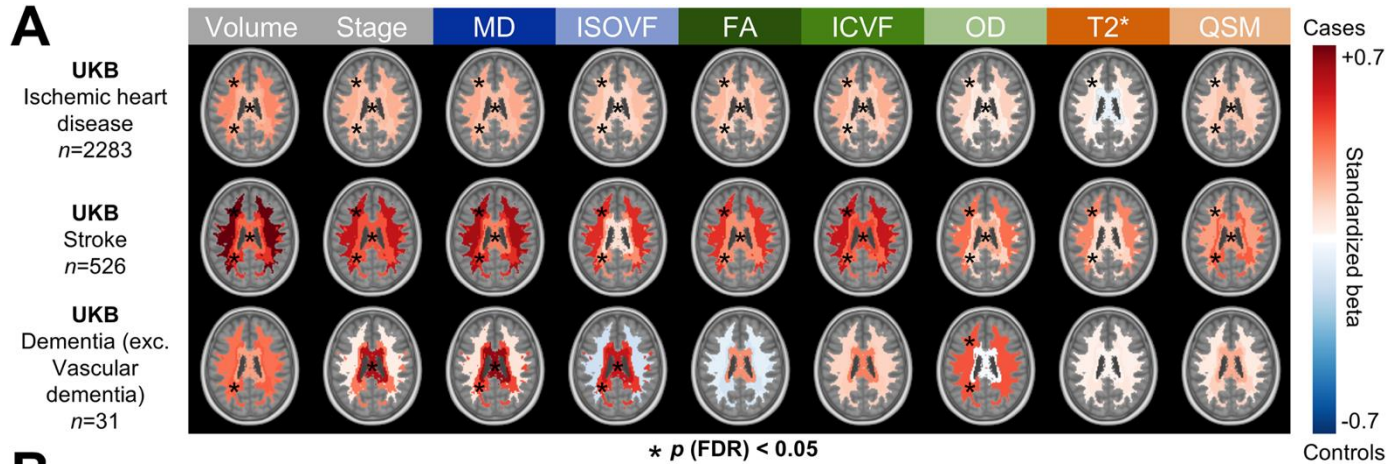
White Matter

White Matter Hyperintensities – An update

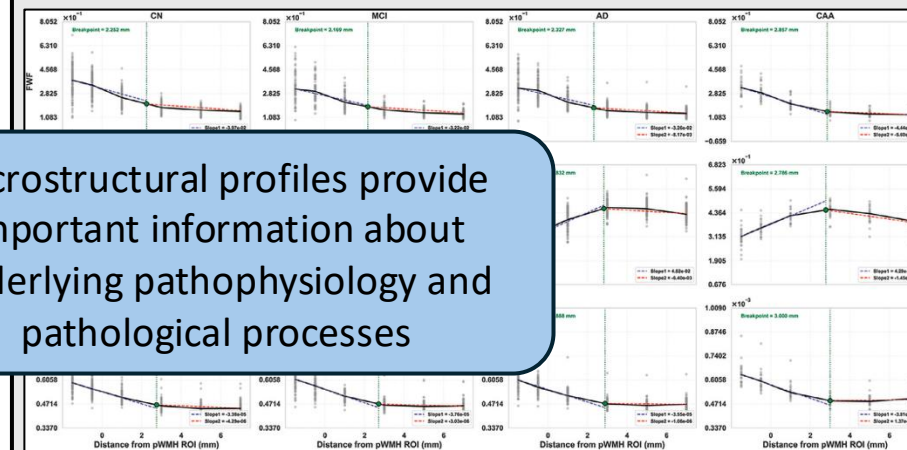
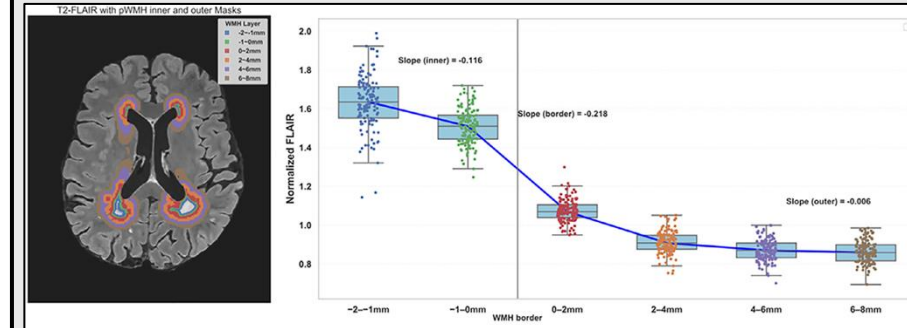


Based on microstructural properties, 3 spatial clusters of pathophysiologically similar WMHs identified in large cohort

Differ in different diseases both in the patterns and in the metrics that are most affected



Perilesional microstructural characterization provide information about underlying pathology:
CAA shows a distinctive pattern of FW changes in perilesional regions with more distal breakpoints and slower plateauing



Microstructural profiles provide important information about underlying pathophysiology and pathological processes



Altschuler et al.
Alz Res & Therapy

Lin et al. Alz & Dem

Yang et al. Alz & Dem

Morillo-Nieto et al. Alz & Dem

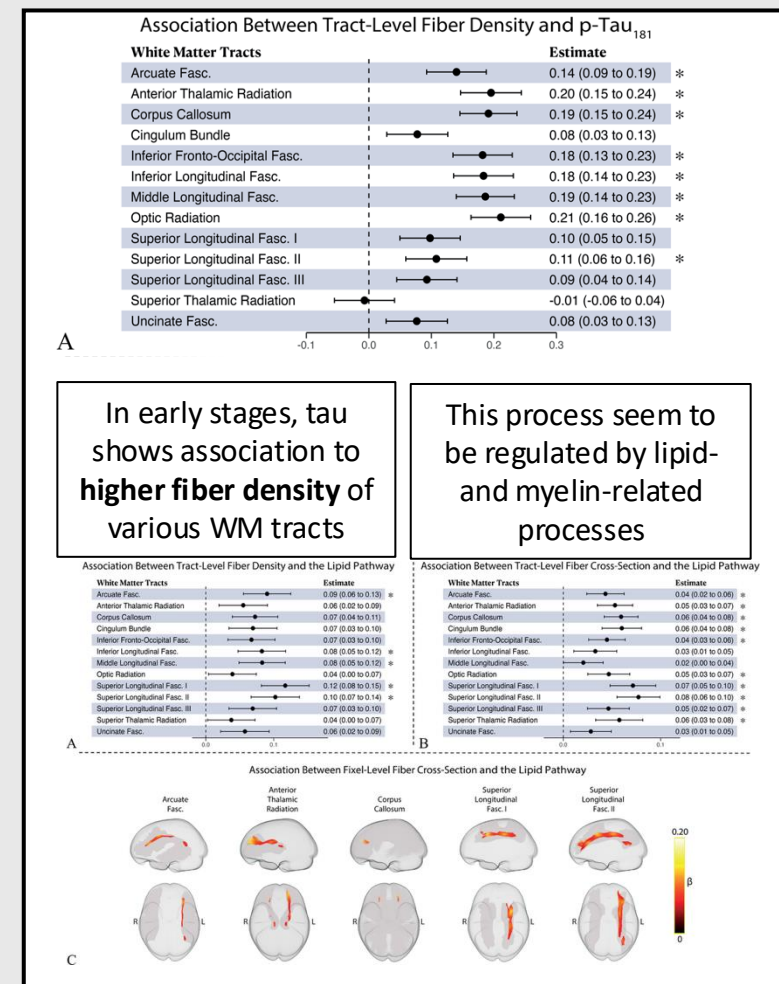
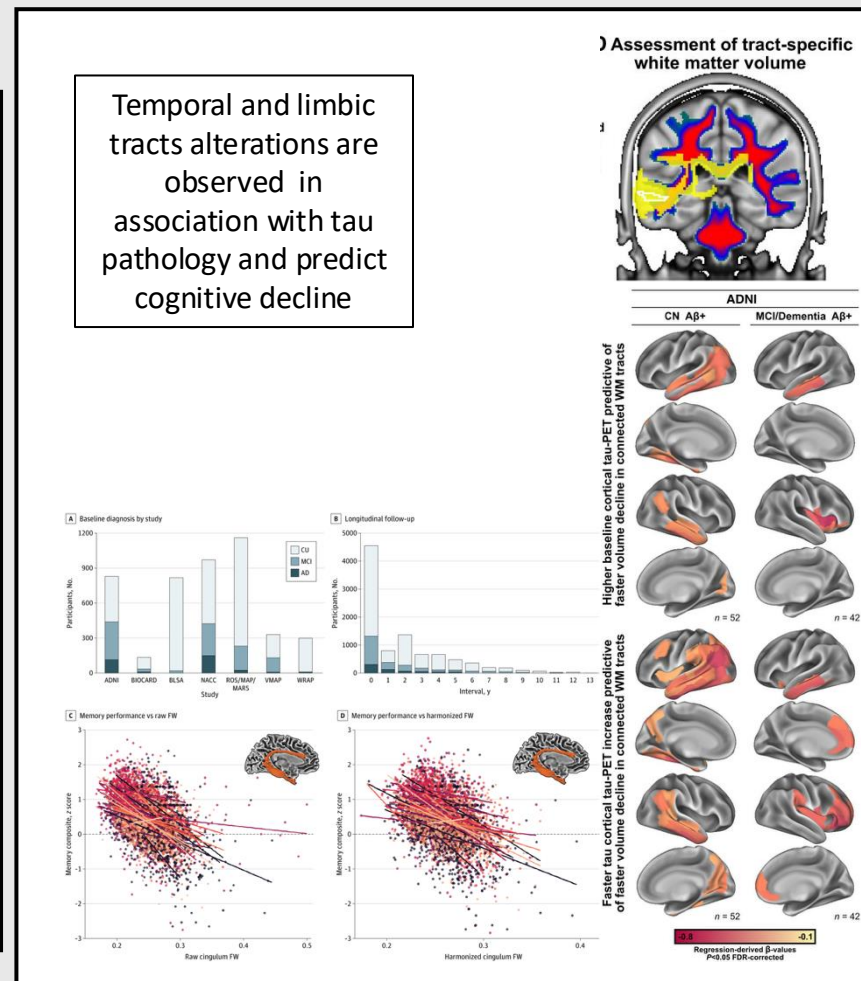
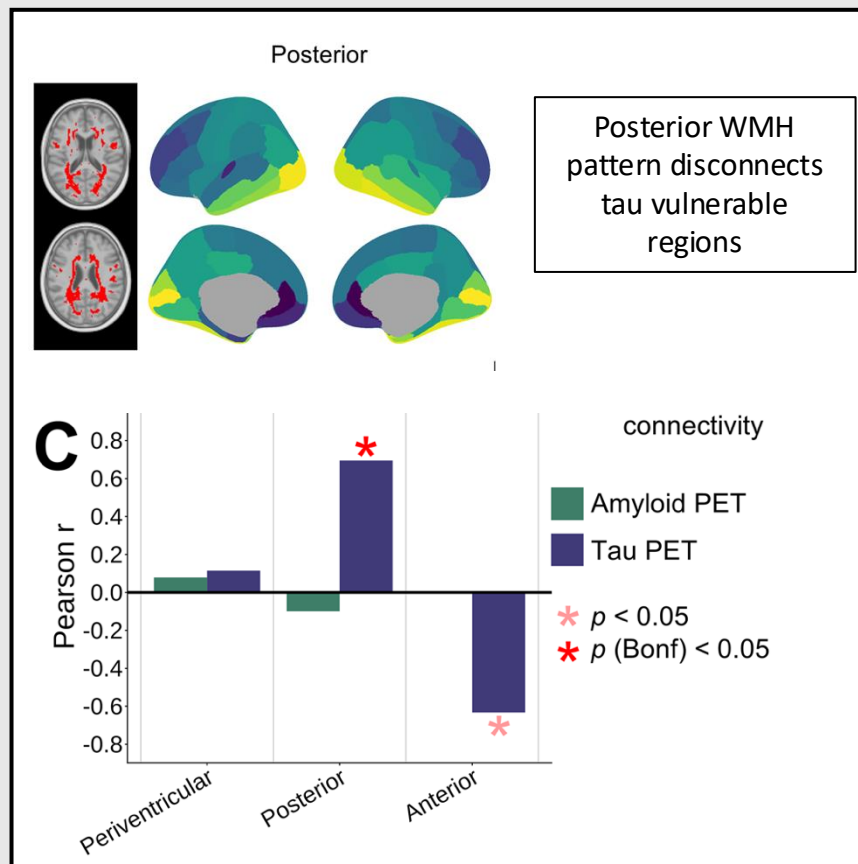
Bachmann et al. Alz Res & Therapy

Parent et al. Nat Comms



White Matter

The mechanistic role of WM in AD



Peter et al.
JAMA Neur

Pescoller et al.
Brain

Tranfa et al. Alz
Res & Therapy

Parent et al. Nat
Comms

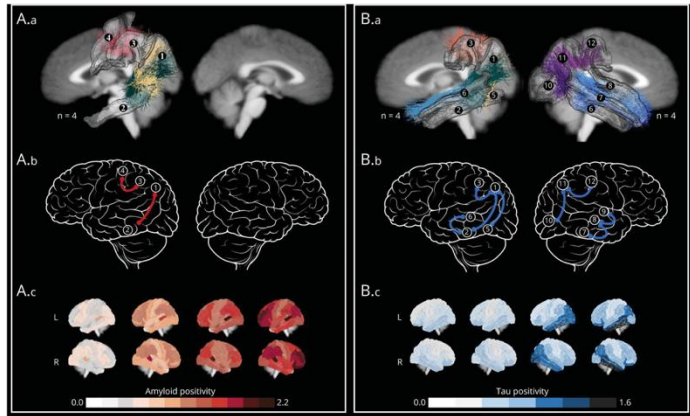


White Matter

The mechanistic role of WM in AD

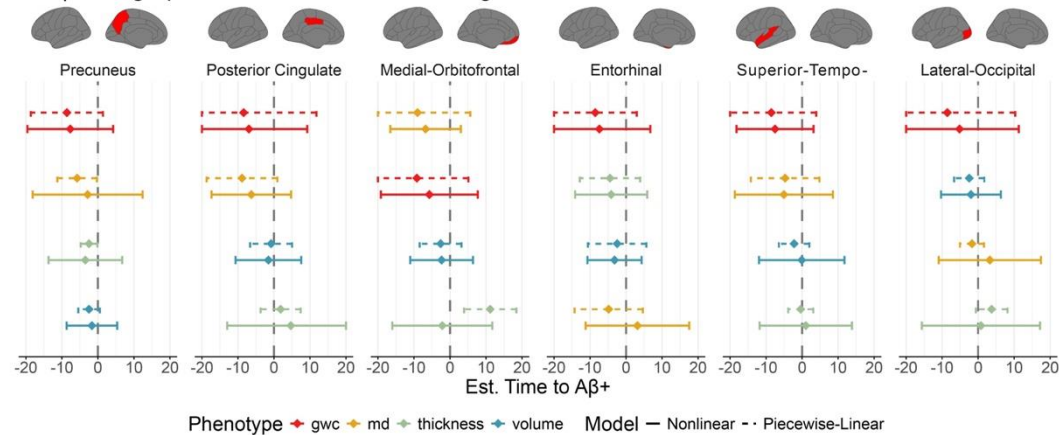
Scratching Beneath the Cortex

Superficial White Matter and Alzheimer Disease

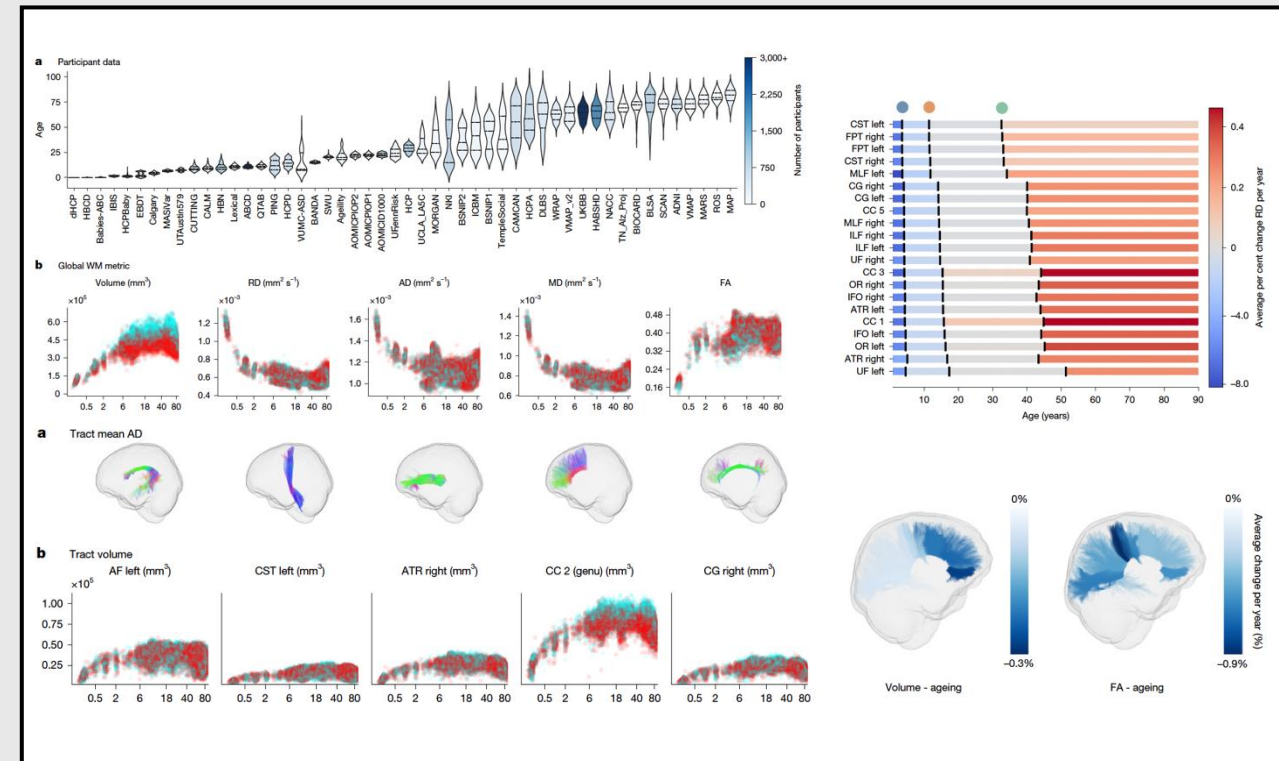


Superficial WM free water and Gray-White matter contrast show early alterations across the AD continuum

G | Change-points across methods and regions



Modelling WM micro and macro structural changes throughout the lifespan



Peter et al.
JAMA Neur

Wang et al.
Neurology

Pescoller et al.
Brain

Tranfa et al. Alz
Res & Therapy

Parent et al. Nat
Comms

Kim et al. Nature

Pieperhoff et al.
Alz & Dem



General AD – Pathophysiology



Cognitive Decline & Intervention



Deep Learning & AI



Connectomics



White Matter



Omics & Biological Underpinnings



Omics & Biological Underpinnings

Understanding the biology of imaging endophenotypes

Understanding Disease Heterogeneity

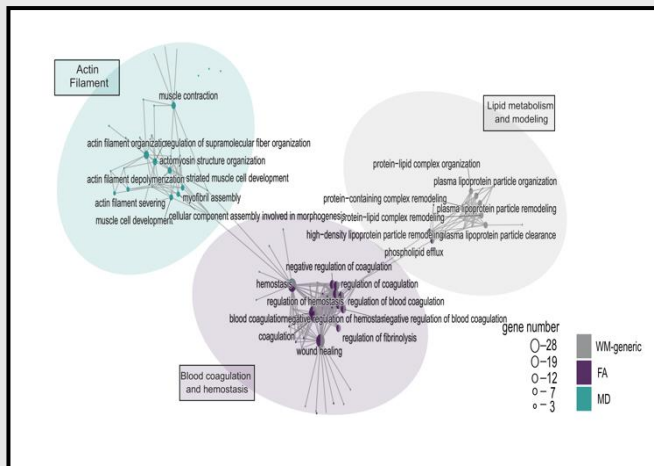


Omics & Biological Underpinnings

Understanding the biology of imaging endophenotypes

Differential and shared proteomics pathways contribute to white matter health and disease

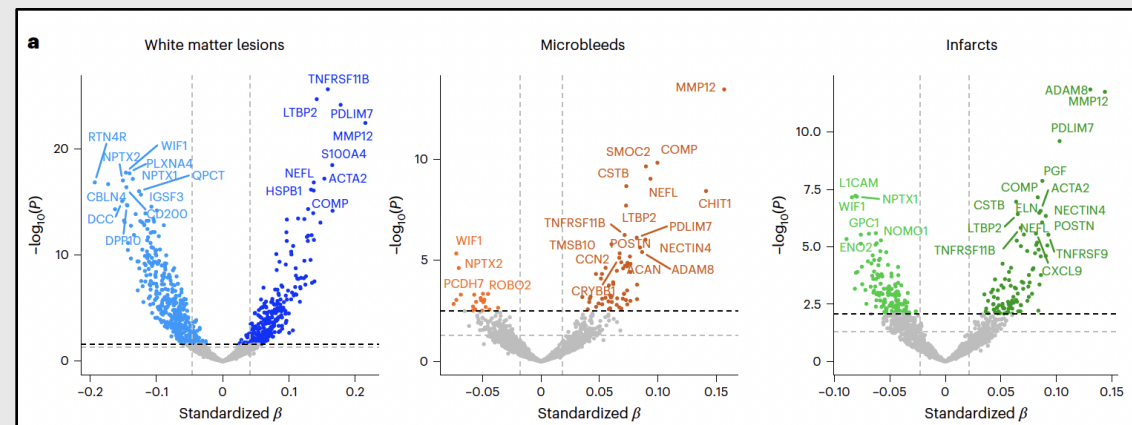
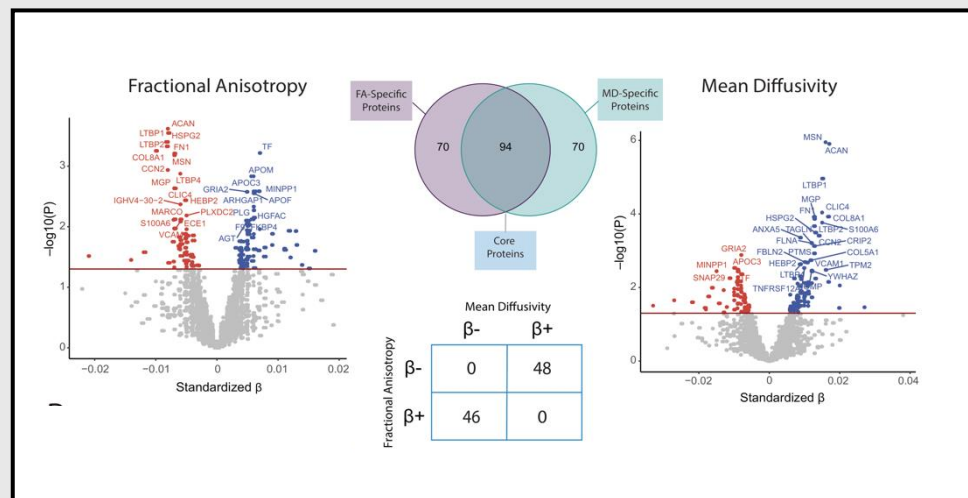
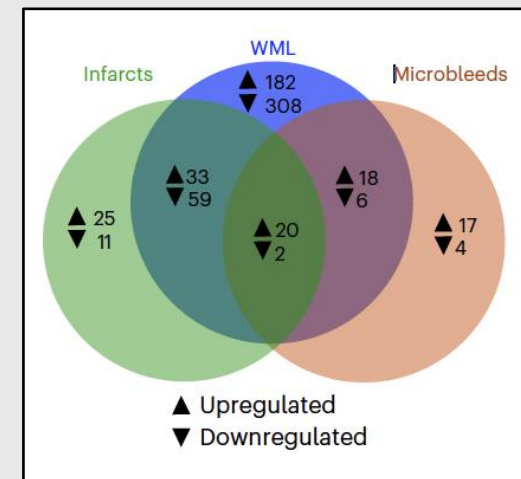
DTI Scalars



DTI metrics are commonly related to lipid pathways, and differentially to vascular (FA) and structural (MD) pathways

Vascular cell dysfunction promotes the appearance of cSVD, while inflammation the progression

CSVD Markers



Sim et al. Alz & Dem



Lorenzini et al. Neurology

Haghighi et al. Alz & Dem: DADM17

Hristovska et al. Nature Aging

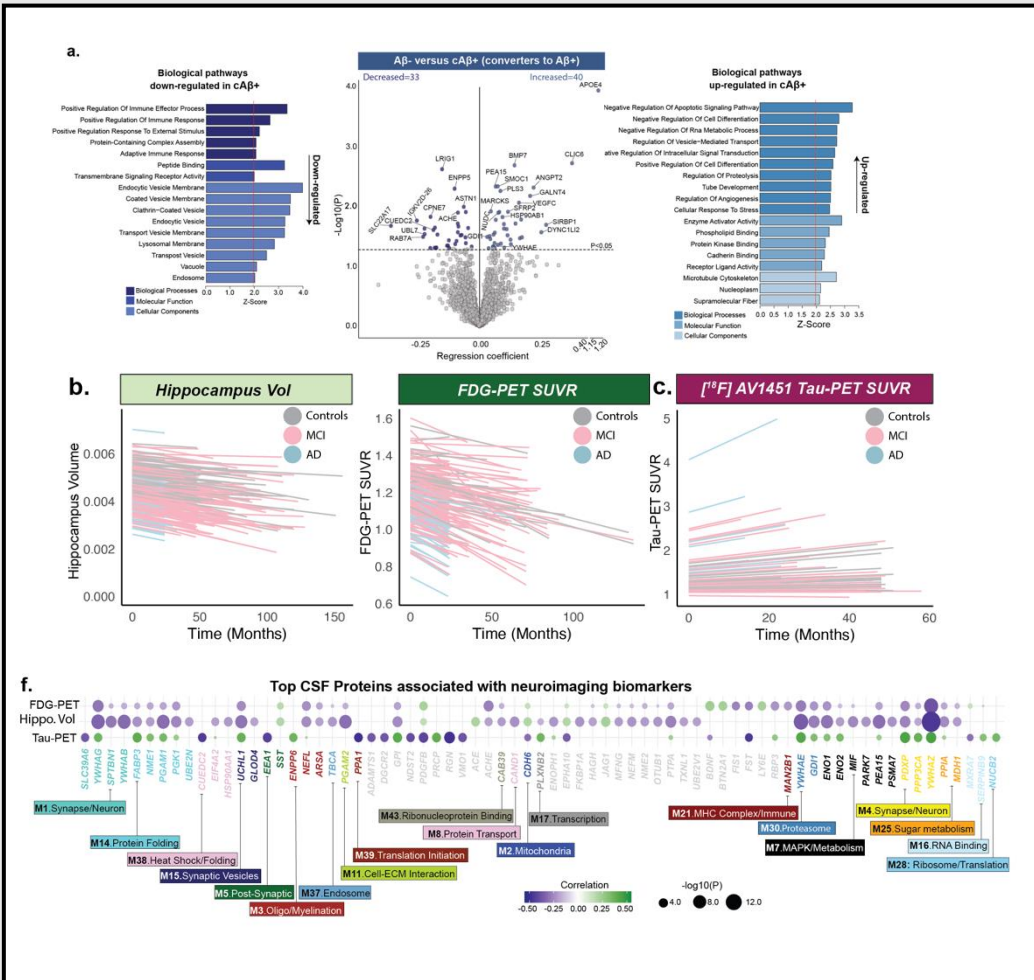


Omics & Biological Underpinnings

Understanding the biology of imaging endophenotypes

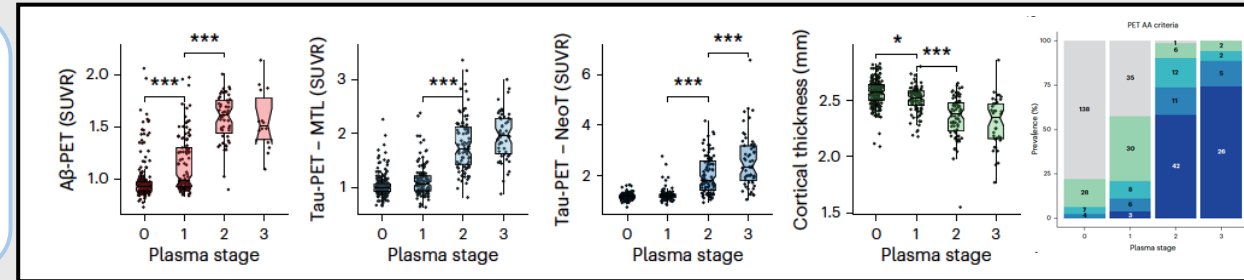
CSF proteomics reveal distinct waves of biological alterations

Plasma P-tau isoforms can specifically track progression in A β and tau PET



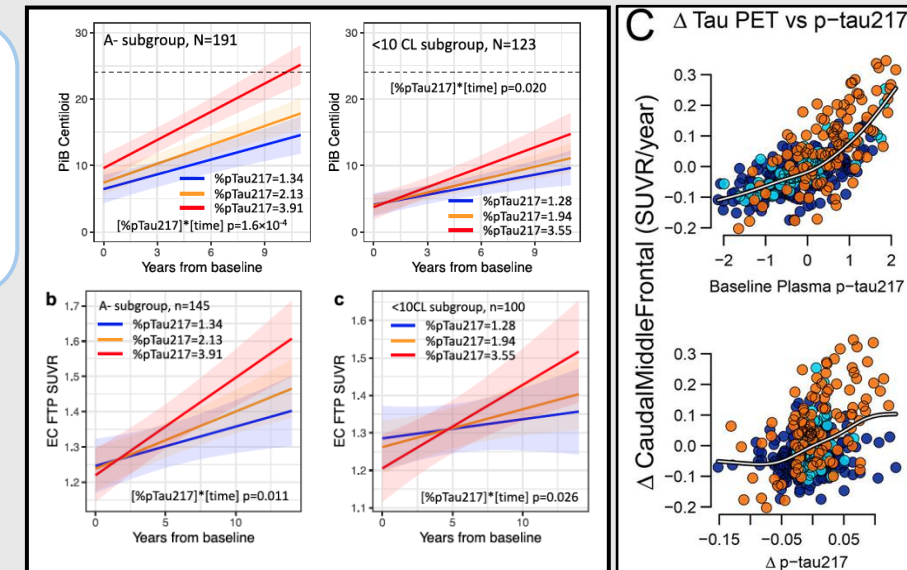
A set of proteins and pathways specifically altered in future A β positive (converters)

Both shared and distinct biological pathways determine longitudinal progression in hypometabolism, hippocampal atrophy and tau PET. Suggesting core and distinct disease-related pathways



Biological stages defined with Plasma p-tau isoforms strongly associated with A β and tau PET

Longitudinally, plasma p-tau predicts amyloid and tau accumulation and it seems to increase before amyloid PET



Pichet Binette et al. Nat Neur

Montoliu-Gaya et al. Nat Aging

Insel et al. Jama Neuro

Yang et al. Nat Comms

Rathore et al. MedRxiv

Labrador-Espinosa Alzh Res & Therapy



(B)

Limbic Atrophy

Correlation Coefficients

SC FC MS

Diffuse Atrophy

Correlation Coefficients

SC FC MS

Revised p-values
Autocorr. p-values

(C)

GO terms for Limbic Atrophy

GO terms for Diffuse Atrophy

(D)

Biological Processes Network

Figure 2: eNDM-predicted vulnerability.

A Observed regional tau (EMCI, LMCI, AD) and eNDM-predicted tau (EMCI, LMCI, AD) across lateral (L) and medial (R) views. Color scales range from 0.000 to 0.005.

B Vulnerability: network-aligned (SV-NA) and vulnerability: network-independent (SV-NI) across lateral (L) and medial (R) views. Color scales range from 0.000 to 0.005.

C Network diagrams for SV-NA and SV-NI. Nodes represent biological processes, and edges represent network connections.

SV-NA Network Diagram:

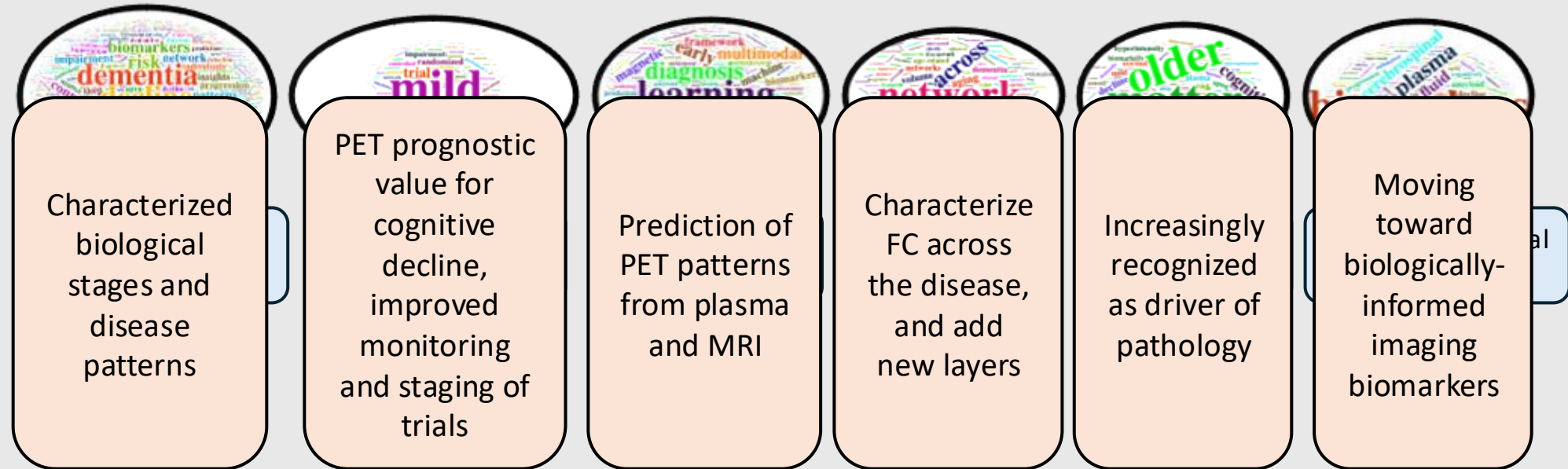
- Neuron death
- Reg. of neuron death
- Pos. reg. of protein modification proc.
- Pos. reg. of cellular protein metabolic proc.
- Reg. of protein modification proc.
- Reg. of phosphate metabolic proc.
- Reg. of phosphorus metabolic proc.
- Pos. reg. of molecular function
- Reg. of cell death
- Positive reg. of cellular protein metabolic proc.
- Reg. of phosphate metabolic proc.
- Reg. of phosphorus metabolic proc.
- Positive reg. of protein modification proc.

SV-NI Network Diagram:

- Prod. of molecular mediator involved in inflammatory response
- Cytokine prod. involved in inflammatory response
- Negative reg. of amyloid precursor protein catabolic proc.
- Chaperone-mediated protein folding
- Protein folding
- Reg. of inclusion body assembly
- Negative reg. of intracellular signal transduction
- Cellular response to lipid
- Negative reg. of amyloid-beta formation
- Cellular response to lipid
- Negative reg. of intracellular signal transduction
- Protein folding
- Negative reg. of amyloid precursor protein catabolic proc.

Boccalini et
al. *EANM*

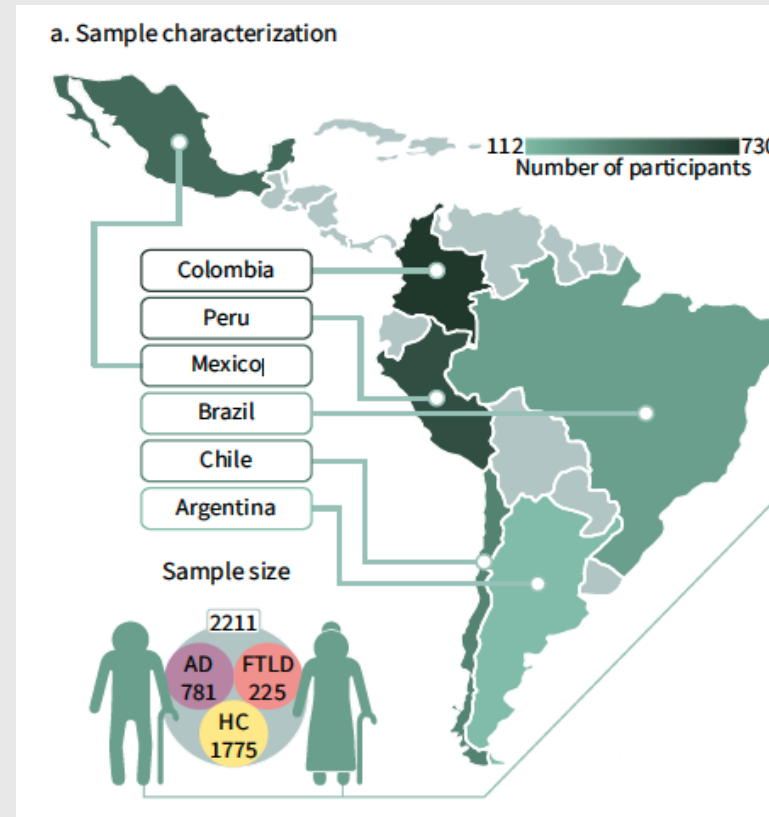
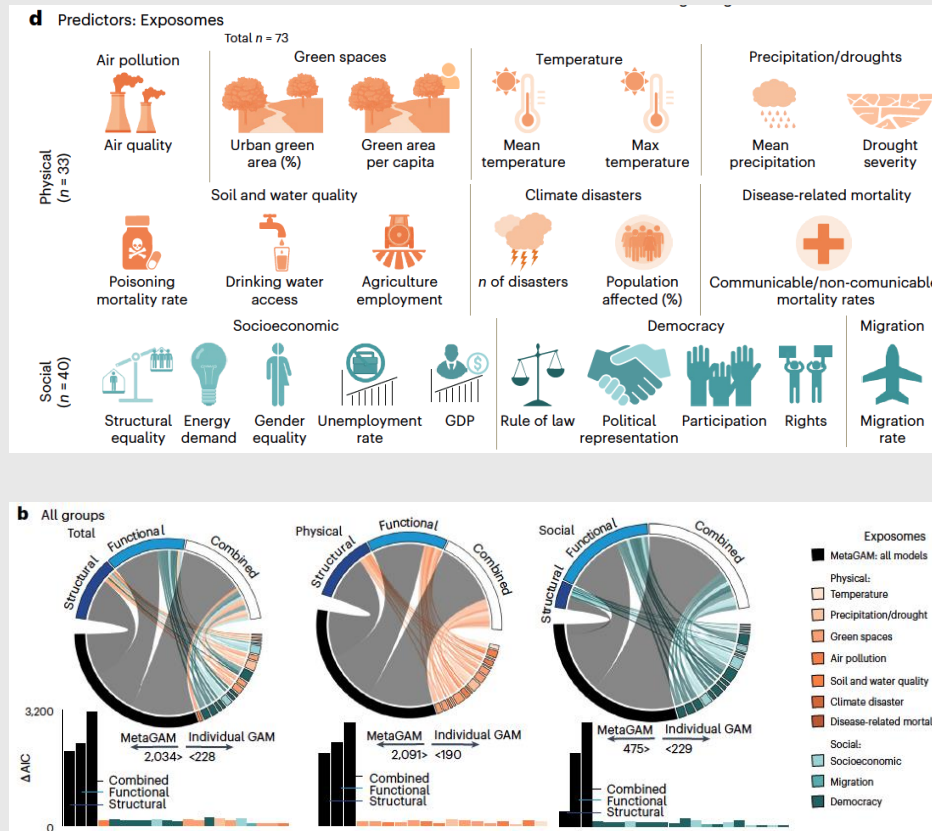
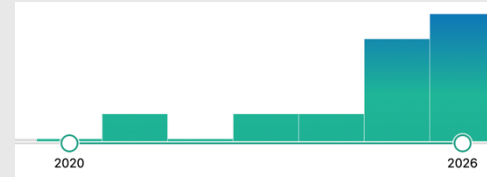
What have we learned?





What's next?

Accelerated Brain Ageing Associated with the **Exposome**



Migeot *al. Nat Comms*

Legaz *al. Nat Medicine*

Genon *al. Nat Neur Reviews*

Sept

Oct

Nov

Dec

Jan

Feb

Mar

Apr

May

Jun

The future!

Thank you for your attention!



Amsterdam UMC

Frederik Barkhof

Alle Meije Wink

Henk Jan Mutsaerts

Betty Tijms

Mario Tranfa

Leonard Pieperhoff

Prithvi Arunachalam

Federico Masserini



UNIVERSITÀ DEGLI STUDI
DI GENOVA

Matteo Pardini

Luca Roccatagliata

Beatrice Orso

Sara Garbarino

Mattia Losa

Stefano Caneva

Questions or comments?

Ask me at the drinks tonight!

Or find me at:



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Luigi.lorenzini@edu.unige.it



AMYPAD

Amyloid imaging to prevent
Alzheimer's Disease

EPAD

European Prevention of
Alzheimer's Dementia Consortium



Funding bodies



Slides of this talk can be
found here: